CG3CG3 — Chapters 3 & 4

de Pedro, Joenell Conrad   Florentino, Ian Miguel
Nacu, Drixyl Reece   Ruelan, Ivan Paul   Seno, Matthew Anakin

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Contents

CHAPTER 3: METHODOLOGY

3.1 Research Design

This study adopts a quantitative, non-experimental research design using secondary data obtained from the Kaggle Developer Survey dataset. The research is descriptive and predictive in nature, aiming to identify factors that significantly influence developer compensation and to construct a reliable predictive model for annual salary.

The analytical approach is grounded in multiple linear regression, a classical statistical method suited for estimating the linear relationship between a continuous dependent variable and multiple independent predictors. Because the data are observational rather than experimentally controlled, the study focuses on identifying statistically significant associations rather than making causal claims.

Model validity is assessed through a comprehensive suite of diagnostic tests covering the core OLS assumptions: normality of residuals, homoscedasticity, independence of errors, and absence of severe multicollinearity. Multiple model specifications are compared using Adjusted R^2, AIC, and BIC to arrive at the most parsimonious and well-fitting final model.

3.2 Data Collection

The dataset used in this study was sourced from Kaggle, a publicly accessible platform for open datasets widely used in academic and data science research. The dataset is a **synthetically generated** developer survey containing 12,000 records and 32 variables designed to simulate realistic developer demographics, compensation, and technology adoption patterns.

Dataset Source: https://www.kaggle.com/datasets/meruvakodandasuraj/developer-survey-2026-skills-salary-and-ai-impact

The dataset was loaded into R using the following command:

Table 1: Table 1. Developer Demographics

respondent_id	survey_year	country	age	gender	education_level	years_of_experience	
R00001	2024	India	43	Man	Some college	22	Fron
R00002	2025	Italy	28	Man	Master's degree	7	Machine
R00003	2021	South Africa	23	Woman	Master's degree	0	Machine
R00004	2023	India	22	Prefer not to say	Bachelor's degree	1	QA
R00005	2025	United States	34	Man	Bachelor's degree	11	Fron
R00006	2021	India	33	Man	Bachelor's degree	15	Fron
R00007	2022	China	29	Man	Bachelor's degree	13	Cl
R00008	2022	India	31	Man	Bachelor's degree	12	Full-
R00009	2025	United States	46	Man	Bachelor's degree	26	Fron
R00010	2020	Indonesia	37	Man	Self-taught / Bootcamp	11	Fron
R00011	2021	Pakistan	20	Prefer not to say	Master's degree	0	Emb
R00012	2026	United States	36	Woman	Master's degree	16	QA
R00013	2022	South Africa	50	Man	Self-taught / Bootcamp	25	Full-
R00014	2025	Belgium	24	Man	Bachelor's degree	0	Fron
R00015	2021	India	21	Man	Master's degree	2	St

Table 2: Table 2. Employment and Salary Information

employment_status	company_size	work_mode	coding_hours_per_week	annual_salary_usd	job_satisfaction	ca
Unemployed	1-10	Hybrid	12	3700	1	
Unemployed	201-1000	Remote	26	9660	3	
Full-time	1001-5000	In-Office	43	26200	4	
Unemployed	10000+	Hybrid	27	0	5	
Full-time	51-200	Remote	25	172100	3	
Full-time	51-200	In-Office	29	14900	4	
Full-time	1001-5000	Hybrid	27	45200	1	
Full-time	51-200	Remote	34	16200	5	
Full-time	201-1000	Hybrid	40	245100	4	
Full-time	1001-5000	In-Office	45	17900	2	
Student	5001-10000	Hybrid	12	0	4	
Full-time	11-50	In-Office	33	211500	3	
Full-time	1-10	Hybrid	35	46500	5	
Full-time	5001-10000	Remote	41	34600	4	

Table 2: Table 2. Employment and Salary Information *(continued)*

employment_status	company_size	work_mode	coding_hours_per_week	annual_salary_usd	job_satisfaction	c
Full-time	201-1000	Hybrid	51	4600	4	

Table 3: Table 3. Technology Stack

primary_os	languages_used	frameworks_used	databases_u
macOS	Swift;JavaScript;Python;HTML/CSS;SQL;Kotlin	Node.js;Ruby on Rails	PostgreSQL;Firebase Realtim
macOS	Python;HTML/CSS;Java	FastAPI	PostgreSQL;SQLite;Mon
Linux	Swift;Python;Java;JavaScript;Bash/Shell;SQL	Spring Boot	DynamoDB
Linux	Python;JavaScript;TypeScript	Express	PostgreSQL;Oracle
Windows	C++;JavaScript;Scala;TypeScript	Django;.NET;Vue.js;Next.js;React	Firebase Realtime DB;SQLit
Windows	JavaScript;C;Python;SQL	React;Node.js;Laravel;Express;Flutter	SQLite;Microsoft SQL Ser
Linux	PHP;Bash/Shell;SQL;Python	Spring Boot;htmx;Tailwind CSS;Bootstrap	MySQL
Windows	C;Kotlin;Bash/Shell;SQL;JavaScript	FastAPI;React	Microsoft SQL
macOS	JavaScript	Flask;Next.js;TensorFlow	Redis;PostgreSQL;Supa
Windows	C#;Rust;HTML/CSS	Vue.js;Bootstrap	DynamoDB;PostgreSQL;Coc
macOS	SQL;JavaScript	Django;Flask;Angular;Next.js	PostgreSQL;Microsoft
macOS	SQL;Go;Python;JavaScript	TensorFlow;Tailwind CSS;NestJS	Firebase Realtime DB
Windows	PHP;HTML/CSS;TypeScript;SQL	Spring Boot;Next.js	Cassandra;Neo4j;MySQ
Windows	Zig;Python;JavaScript;C++	Angular	SQLite;Postgre
Windows	PHP	Node.js;jQuery;FastAPI;Spring Boot	Microsoft SQL Server;Redis;

Table 4: Table 4. AI Usage and Perception

ai_tools_used	ai_usage_frequency	ai_productivity_boost_pct	ai_sentiment	ai_job_threat_perceptio
	Weekly	14	Somewhat positive	Minimal threat
	Weekly	8	Somewhat positive	Minimal threat
ChatGPT	Rarely	1	Very negative	Significant threat
	Multiple times daily	61	Neutral	Minimal threat
	Weekly	15	Neutral	Moderate threat
	Daily	36	Very positive	Significant threat
	Monthly	8	Somewhat positive	Minimal threat
	Multiple times daily	16	Neutral	No threat
ChatGPT	Weekly	26	Somewhat positive	Will replace most devs
ChatGPT;Cursor AI	Weekly	19	Very negative	Significant threat
Cursor AI;ChatGPT;Claude	Never	9	Somewhat positive	Moderate threat
ChatGPT	Multiple times daily	36	Very positive	No threat
	Monthly	4	Very positive	Minimal threat
Claude Code;Perplexity	Daily	7	Very positive	Moderate threat
	Multiple times daily	45	Very positive	Minimal threat

Table 5: Table 5. Career Growth and Preferences

primary_learning_source	open_source_contributor	looking_for_job	remote_work_preference_1to5	burnout_freq
Online courses	1	Actively looking	2	Sometimes
Stack Overflow	1	Not looking	4	Often
YouTube	0	Open to offers	5	Often
Stack Overflow	1	Open to offers	3	Rarely
Official docs	0	Open to offers	1	Sometimes
Stack Overflow	0	Not looking	3	Sometimes
Official docs	0	Not looking	5	Often
Books	1	Not looking	5	Often
YouTube	0	Not looking	5	Often
ChatGPT/AI	1	Actively looking	4	Rarely
Colleagues	0	Not looking	2	Always
Stack Overflow	0	Not looking	5	Rarely
Blogs	1	Open to offers	3	Always
ChatGPT/AI	1	Open to offers	4	Sometimes
Official docs	0	Open to offers	4	Often

An initial inspection confirmed that only one column, `ai_tools_used`, contained missing values (approximately 19.8% of rows). All other columns were complete prior to imputation. The `ai_tools_used` column was excluded from the model as it is a semicolon-delimited multi-value free-text field not directly suitable for regression encoding.

3.3 Variables Description

The study models annual developer salary as the dependent variable, with a comprehensive set of demographic, employment, educational, and behavioral predictors serving as independent variables.

3.3.1 Dependent Variable

The dependent variable is `log_salary`, defined as the natural logarithm of `annual_salary_usd`. A log transformation was applied because raw salary distributions are strongly right-skewed; the transformation stabilizes variance across fitted values and brings the distribution closer to normality, both of which are assumptions of OLS regression. Respondents with `annual_salary_usd $\leq$ 0` were excluded from the analysis as these entries reflect unemployed or unreported salary statuses rather than true compensation values.

3.3.2 Independent Variables

The independent variables include all numeric, ordinal, and nominal features available in the cleaned dataset after excluding ID columns, multi-value text fields, and the target variable.

Table 3.1 presents the full set of predictors, their measurement type, and how each was encoded for regression.

Variable	Type	Encoding / Notes
years_of_experience	Continuous numeric	Raw numeric; direct linear predictor of salary
age	Continuous numeric	Raw numeric
coding_hours_per_week	Continuous numeric	Raw numeric
ai_productivity_boost_pct	Continuous numeric	Raw numeric; self-reported % boost from AI tools
remote_work_preference_1to5	Ordinal numeric	Raw numeric (1 = fully on-site, 5 = fully remote)
open_source_contributor	Binary numeric	Raw numeric (0/1)
education_level	Ordered factor	7-level ordered factor from High school to Doctorate (PhD)
company_size	Ordered factor	7-level ordered factor from 1–10 to 10000+ employees
burnout_frequency	Ordered factor	5-level ordered factor: Never to Always
ai_usage_frequency	Ordered factor	6-level ordered factor: Never to Multiple times daily
role	Nominal factor	Unordered; each level gets a dummy coefficient vs. baseline
employment_status	Nominal factor	Unordered; e.g., Full-time, Part-time, Freelance
work_mode	Nominal factor	Unordered; e.g., Remote, Hybrid, In-Office
gender	Nominal factor	Unordered
primary_os	Nominal factor	Unordered; e.g., Windows, macOS, Linux
ai_sentiment	Nominal factor	Unordered; attitude toward AI in the workplace
ai_job_threat_perception	Nominal factor	Unordered; perceived job threat level from AI
primary_learning_source	Nominal factor	Unordered; e.g., Official docs, Stack Overflow, YouTube

Variable	Type	Encoding / Notes
looking_for_job	Nominal factor	Unordered; current job-seeking status
country	Nominal factor	Unordered; country of employment

Table 3.1. Independent Variables Used in the Multiple Linear Regression Model

3.4 Data Analysis Techniques

3.4.1 Descriptive Statistics

Exploratory data analysis was conducted prior to modeling to understand variable distributions and guide preprocessing decisions. Specific procedures included:

- Histograms of raw and log-transformed salary to justify the log transformation
- Pearson correlation matrix of all numeric predictors against `log_salary` to screen for linear relationships
- Summary statistics (mean, median, standard deviation, quartiles) via `summary()` in R
- Frequency tables for categorical variables to detect dominant categories and inform mode imputation
- Missing value audit to quantify and locate all NA entries before imputation

Numeric variables with missing values were imputed using the column median, and categorical variables were imputed using the column mode, both strategies chosen to preserve distributional properties without introducing bias from mean substitution in skewed data.

3.4.2 Inferential Statistics

The primary analytical method is multiple linear regression (MLR) estimated by Ordinary Least Squares (OLS), implemented using the `lm()` function in R. The regression equation takes the general form:

$$\log(\text{salary}) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$

where X_1 through X_k represent the independent variables listed in Table 3.1, β_0 is the intercept, $\beta_1 \ldots \beta_k$ are regression coefficients, and ε is the error term. All hypothesis tests are conducted at the $\alpha = 0.05$ significance level.

Significance Testing

- Individual predictor significance was assessed via t-tests on each regression coefficient, with p-values reported from `summary(model)`
- Overall predictor group significance was assessed via F-tests using `anova(model)`
- Predictors were classified as significant ($p < 0.05$) or non-significant ($p \geq 0.05$) and ranked accordingly

Variable Selection

To identify the most parsimonious model, several variable selection strategies were applied and compared:

- Manual stepwise elimination by p-value, producing Models v2, v3, and v4 by progressively removing non-significant predictors
- Backward elimination using `drop1()` with F-tests to identify the weakest contributor at each step
- Bidirectional stepwise AIC selection using `stepAIC(direction = "both")` from the MASS package
- Stepwise BIC selection using `stepAIC()` with `k = log(n)` to apply a stronger complexity penalty
- A theory-driven reduced model retaining only substantively meaningful predictors

All candidate models were evaluated and ranked on Adjusted R^2, AIC, and BIC. The full model was retained as the final model based on this comparison.

Model Assumption Diagnostics

The following formal tests and visual diagnostics were applied to the final model to validate the OLS assumptions:

Assumption	Test / Method	Tool in R
Normality of residuals	Shapiro-Wilk test; Kolmogorov-Smirnov test; Q-Q plot; residual histogram	`shapiro.test()`, `ks.test()`, `qqnorm()`
Homoscedasticity	Breusch-Pagan test; Residuals vs. Fitted plot; Scale-Location plot	`bptest()` from lmtest
Independence of errors	Durbin-Watson test (acceptable range: 1.5 – 2.5)	`dwtest()` from lmtest
Multicollinearity	Variance Inflation Factor (VIF); flag VIF > 5 (moderate) and VIF > 10 (severe)	`vif()` from car

Assumption	Test / Method	Tool in R
Influential observations	Cook's Distance with threshold 4/n; Leverage (hat values) with threshold 2p/n	`cooks.distance()`, `hatvalues()`

Table 3.2. OLS Assumption Diagnostics Applied to the Final Model

In summary, this chapter described a rigorous multiple linear regression pipeline implemented in R, covering data sourcing from Kaggle, preprocessing and imputation, variable operationalization, model fitting, variable selection across multiple strategies, and a comprehensive set of assumption diagnostics. The combination of formal statistical tests and visual diagnostics ensures that the final model is both statistically valid and interpretable.

CHAPTER 4: RESULTS AND DISCUSSION

This chapter presents the results of the multiple linear regression analysis conducted on the Developer Survey 2026 dataset. The findings are organized into four sections: descriptive statistics summarizing the dataset, visualizations of key variable distributions and relationships, model outputs from all candidate regression specifications, and a discussion interpreting the significant predictors and diagnostic results.

4.1 Descriptive Statistics

Descriptive statistics were computed on the cleaned dataset (`data_clean`) using the `summary()` function in R. After removing rows with zero or missing salary values, the final dataset consisted of 11,445 observations across all retained variables.

Table 4.1 presents the descriptive statistics for the continuous variables in the cleaned dataset.

Variable	Min	Median	Mean	Max
survey_year	2020	2024	2023	2026
age	18.0	31.0	31.3	60.0
years_of_experience	0.000	8.000	9.389	41.000
coding_hours_per_week	5.00	33.00	32.89	67.00
annual_salary_usd	2,000	38,000	58,625	368,700
job_satisfaction	1.000	4.000	3.602	5.000
career_satisfaction	1.000	4.000	3.748	5.000
ai_productivity_boost_pct	0.00	23.00	24.46	80.00

Variable	Min	Median	Mean	Max
open_source_contributor	0.000	0.000	0.345	1.000
remote_work_preference_1to5	1.000	4.000	4.056	5.000
log_salary	7.601	10.545	10.509	12.818

Table 4.1a. Descriptive Statistics — Continuous Variables

Variable	Most Frequent	Count	2nd Most Frequent	Count
country	United States	2,067	India	1,611
gender	Man	10,025	Woman	970
education_level	Bachelor's degree	4,878	Master's degree	3,194
role	Full-Stack Developer	2,564	Back-End Developer	1,701
employment_status	Full-time	9,325	Freelance	1,008
company_size	201-1000	2,233	51-200	2,126
work_mode	Hybrid	4,597	Remote	4,406
primary_os	Windows	4,761	macOS	3,706
ai_usage_frequency	Multiple times daily	3,154	Daily	2,868
ai_sentiment	Somewhat positive	4,063	Neutral	2,553
ai_job_threat_perception	Minimal threat	3,410	Moderate threat	3,221
primary_learning_source	Official docs	2,292	Stack Overflow	2,058
looking_for_job	Open to offers	4,549	Not looking	4,351
burnout_frequency	Sometimes	3,999	Often	2,906

Table 4.1b. Descriptive Statistics — Categorical Variables (Top 2 Categories)

4.2 Visualizations

Visual exploration of the data was conducted using `ggplot2` in R. The following plots were generated to understand the distributions of key variables and their relationships with the log-transformed salary outcome.

4.2.1 Salary Distribution

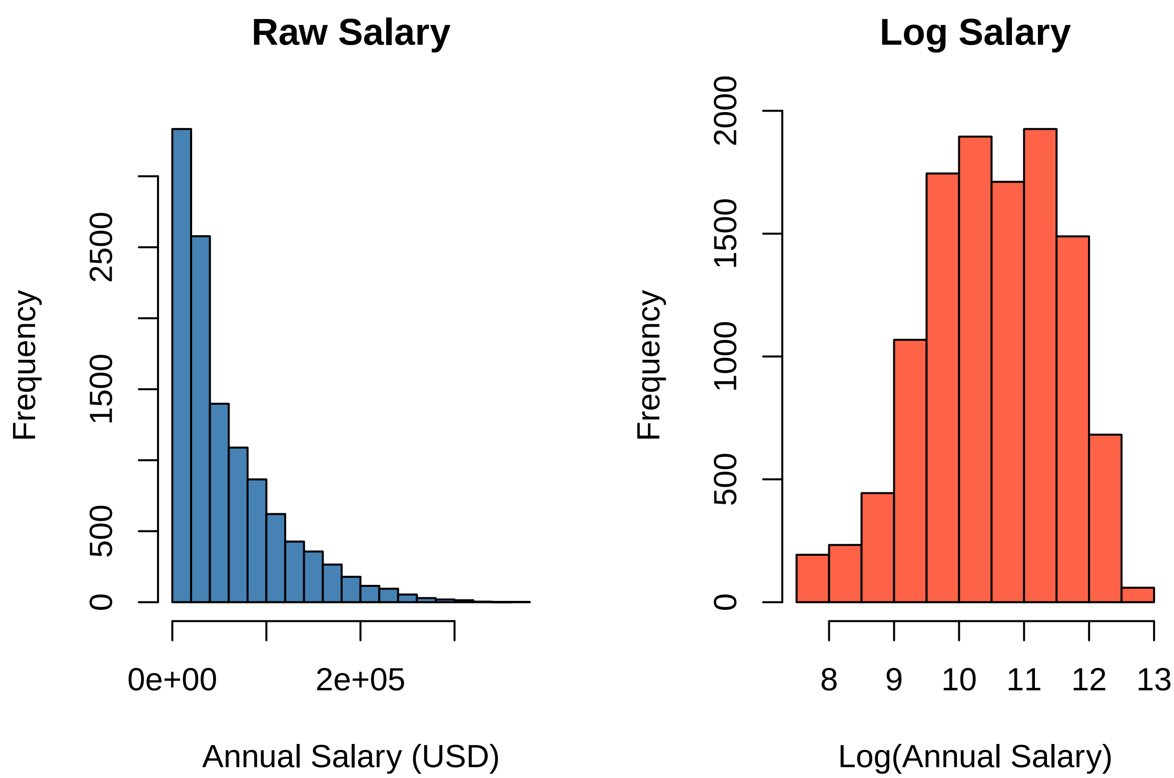

Figure 1: Figure 4.1. Raw vs. Log-Transformed Salary Distribution

The raw salary distribution exhibited strong right skewness, with a long upper tail driven by high-earning roles and senior developers. After applying the natural log transformation, the distribution became approximately symmetric and bell-shaped, confirming that the log transformation was appropriate for meeting the normality assumptions of OLS regression.

4.2.2 Log Salary by Role

Figure 4.2 presents the distribution of log-transformed salary across developer roles. Engineering Manager and DevOps/SRE consistently occupy the highest salary range, with median log salaries approaching 11.0 and above, reflecting strong market demand for infrastructure and leadership roles. Security Engineer and Machine Learning Engineer follow closely, both showing relatively compact interquartile ranges suggesting more consistent compensation within those roles.

Mid-tier roles such as Data Engineer, Data Scientist, Cloud Engineer, and Freelancer/Consultant cluster around a median log salary of approximately 10.5, aligning with the overall dataset median. Back-End, Mobile, Full-Stack, and Front-End Developers occupy

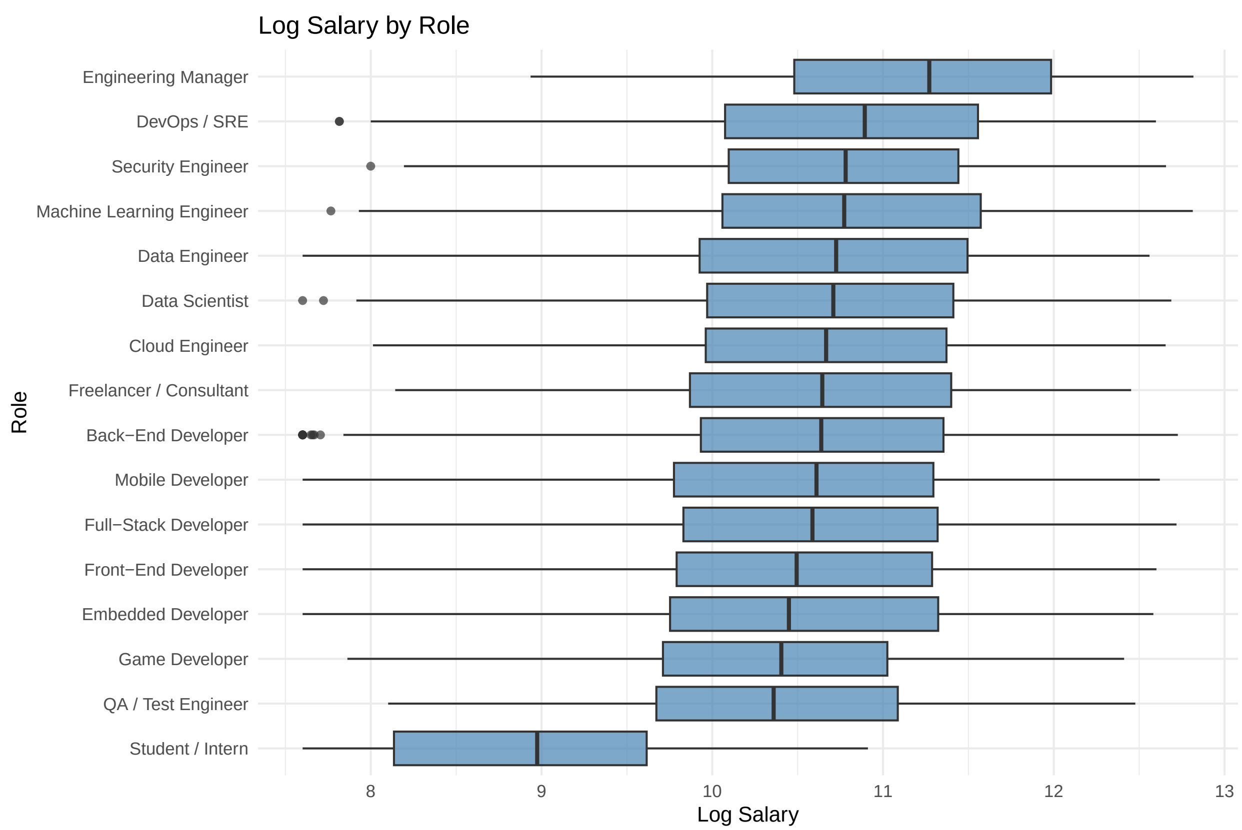

Figure 2: Figure 4.2. Log Salary by Developer Role

a similar range but with wider spreads, indicating greater variability in compensation likely driven by geography, company size, and experience level.

The lowest salary distributions belong to Game Developer, Embedded Developer, QA/Test Engineer, and notably Student/Intern — the latter showing a distinctly lower median of approximately 8.5 and a compressed range consistent with entry-level or unpaid compensation structures. The clear separation between Student/Intern and all other roles confirms that employment status and role type are among the most influential structural determinants of developer salary, which is further supported by their statistical significance in the regression model.

4.2.3 Log Salary by Employment Status

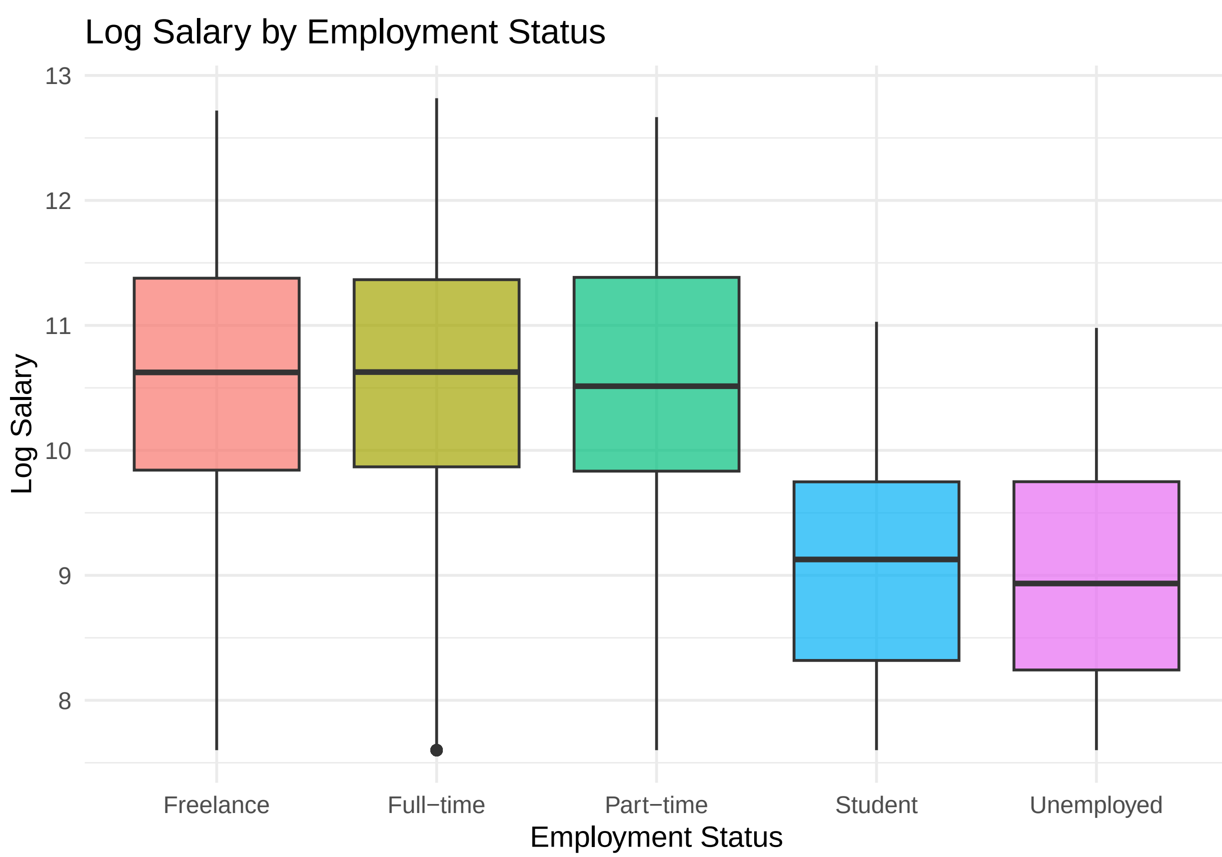

Figure 3: Figure 4.3. Log Salary by Employment Status

Figure 4.3 displays the distribution of log-transformed salary across the five employment status categories. Freelance, Full-time, and Part-time developers share nearly identical median log salaries of approximately 10.7–10.8, with similarly wide interquartile ranges spanning roughly 10.0 to 11.5, suggesting that compensation levels are broadly comparable across these active employment types. In stark contrast, Student and Unemployed respondents show substantially lower median log salaries of approximately 9.0, with compressed distributions and notably less upward spread, reflecting the structural absence of full professional compensation in these categories. The single outlier visible below the Full-time whisker indicates a small number of full-time developers reporting unusually low salaries. The sharp divide between the three active employment categories and the Student/Unemployed groups reinforces the regression finding that employment status is one of the most statistically significant predictors of developer salary.

4.2.4 Log Salary vs. Years of Experience

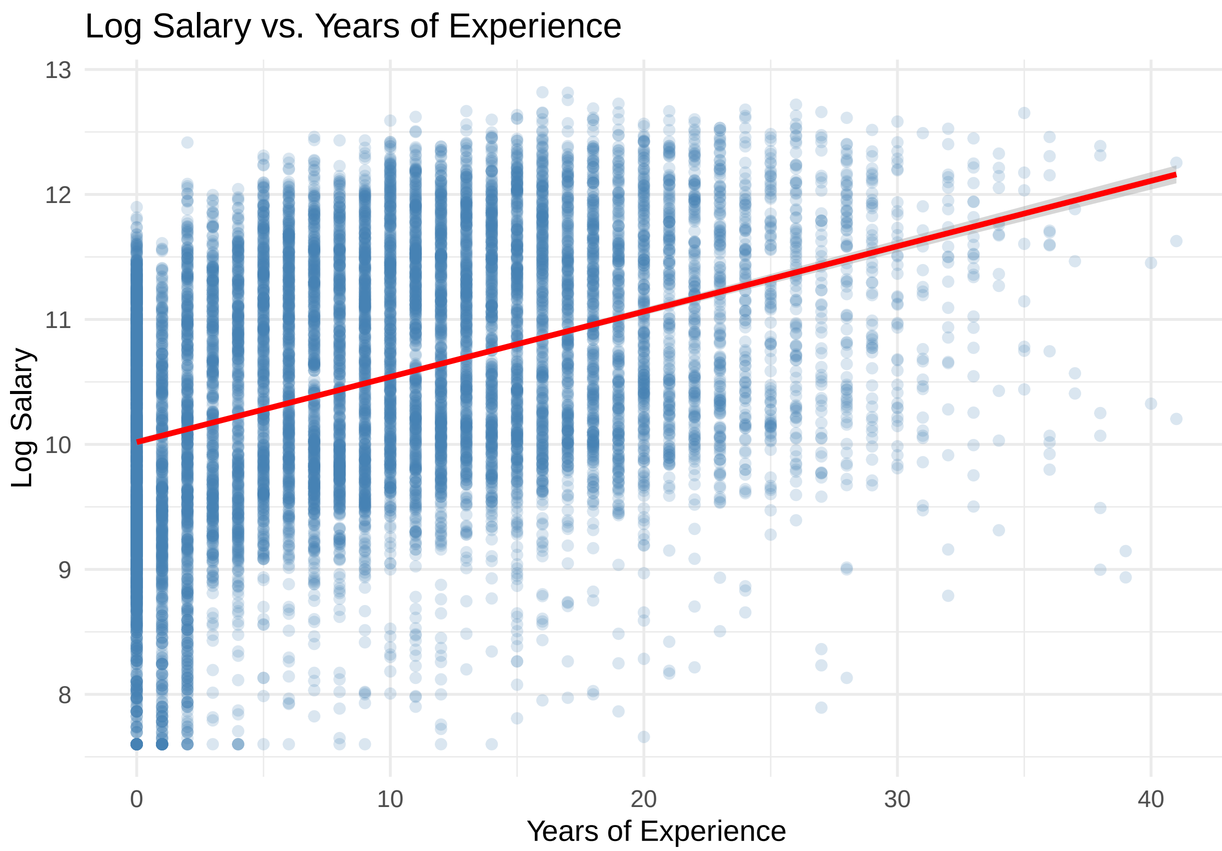

Figure 4: Figure 4.4. Log Salary vs. Years of Experience

Figure 4.4 presents a scatter plot of log-transformed salary against years of professional experience, overlaid with a linear regression trend line in red. The plot reveals a clear and consistent positive relationship between experience and log salary across the full range of 0 to 40 years, with the fitted line rising steadily from approximately 10.0 at entry level to 12.2 at 40 years of experience. The vertical banding visible throughout the plot reflects the discrete integer nature of the years of experience variable. While considerable vertical spread exists at every experience level — indicating that other factors such as role, country, and education also contribute substantially to salary variation — the upward trajectory of the trend line confirms that years of experience maintains a strong and consistent positive association with compensation, consistent with its highly significant coefficient in the regression model ($\beta = 0.0372$, $p < 2e\text{-}16$).

4.3 Model Results

Six model specifications were estimated and compared. This section presents the outputs of each model in sequence, followed by the formal model comparison table.

4.3.1 Variable Selection

Full Model (model_full)

The full model includes all 16 predictors available in `data_clean`.

```r
model_full <- lm(
  log_salary ~ years_of_experience + age + coding_hours_per_week +
    education_level + role + employment_status + company_size +
    work_mode + gender + primary_os + ai_productivity_boost_pct +
    ai_usage_frequency + burnout_frequency + remote_work_preference_1to5 +
    country + open_source_contributor,
  data = data_clean
)

summary(model_full)
```

```
##
## Call:
## lm(formula = log_salary ~ years_of_experience + age + coding_hours_per_week +
##     education_level + role + employment_status + company_size +
##     work_mode + gender + primary_os + ai_productivity_boost_pct +
##     ai_usage_frequency + burnout_frequency + remote_work_preference_1to5 +
##     country + open_source_contributor, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.07264 -0.19443  0.00423  0.19910  1.57541
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  9.1220423  0.0395502 230.645  < 2e-16 ***
## years_of_experience          0.0364405  0.0007850  46.419  < 2e-16 ***
## age                          0.0051502  0.0007936   6.490 8.97e-11 ***
## coding_hours_per_week       -0.0008088  0.0003302  -2.450 0.014312 *
## education_level.L            0.3023787  0.0146232  20.678  < 2e-16 ***
## education_level.Q            0.0065115  0.0138730   0.469 0.638816
## education_level.C            0.0133887  0.0125183   1.070 0.284851
## education_level^4           -0.0150140  0.0106393  -1.411 0.158218
```

```
## education_level^5               0.0079708  0.0099920   0.798 0.425047
## education_level^6              -0.0159580  0.0086094  -1.854 0.063828 .
## roleCloud Engineer              0.0994989  0.0181616   5.479 4.38e-08 ***
## roleData Engineer               0.0525632  0.0151615   3.467 0.000528 ***
## roleData Scientist              0.0410703  0.0134789   3.047 0.002317 **
## roleDevOps / SRE                0.0944514  0.0132448   7.131 1.06e-12 ***
## roleEmbedded Developer          0.0017115  0.0215246   0.080 0.936627
## roleEngineering Manager         0.1996415  0.0178426  11.189  < 2e-16 ***
## roleFreelancer / Consultant     0.0012661  0.0153388   0.083 0.934216
## roleFront-End Developer        -0.1074213  0.0106072 -10.127  < 2e-16 ***
## roleFull-Stack Developer       -0.0493855  0.0091760  -5.382 7.51e-08 ***
## roleGame Developer             -0.1547076  0.0204150  -7.578 3.78e-14 ***
## roleMachine Learning Engineer   0.1394569  0.0140640   9.916  < 2e-16 ***
## roleMobile Developer           -0.0545827  0.0142087  -3.842 0.000123 ***
## roleQA / Test Engineer         -0.2355854  0.0176843 -13.322  < 2e-16 ***
## roleSecurity Engineer           0.1095602  0.0205784   5.324 1.03e-07 ***
## roleStudent / Intern           -1.3698980  0.0137988 -99.276  < 2e-16 ***
## employment_statusFull-time      0.0075404  0.0100400   0.751 0.452649
## employment_statusPart-time      0.0049953  0.0152670   0.327 0.743525
## employment_statusStudent       -1.5858006  0.0185588 -85.447  < 2e-16 ***
## employment_statusUnemployed    -1.5988306  0.0235188 -67.981  < 2e-16 ***
## company_size.L                  0.0005236  0.0079830   0.066 0.947706
## company_size.Q                  0.0097415  0.0073254   1.330 0.183599
## company_size.C                 -0.0035431  0.0077497  -0.457 0.647540
## company_size^4                  0.0039750  0.0079671   0.499 0.617840
## company_size^5                  0.0095829  0.0075203   1.274 0.202592
## company_size^6                 -0.0027050  0.0067018  -0.404 0.686494
## work_modeIn-Office              0.0025645  0.0073452   0.349 0.726994
## work_modeRemote                -0.0075108  0.0061890  -1.214 0.224936
## genderNon-binary               -0.0173878  0.0202282  -0.860 0.390039
## genderPrefer not to say         0.0426571  0.0193541   2.204 0.027542 *
## genderWoman                    -0.0108821  0.0098750  -1.102 0.270491
## primary_osmacOS                 0.0062107  0.0072251   0.860 0.390029
## primary_osWindows              -0.0004203  0.0068641  -0.061 0.951179
## ai_productivity_boost_pct      -0.0001665  0.0002245  -0.742 0.458385
## ai_usage_frequency.L            0.0103565  0.0098926   1.047 0.295168
## ai_usage_frequency.Q           -0.0052350  0.0075310  -0.695 0.486990
## ai_usage_frequency.C           -0.0048955  0.0075187  -0.651 0.514990
## ai_usage_frequency^4            0.0146814  0.0074533   1.970 0.048886 *
## ai_usage_frequency^5           -0.0087918  0.0076975  -1.142 0.253408
## burnout_frequency.L             0.0070305  0.0084390   0.833 0.404808
## burnout_frequency.Q             0.0067552  0.0075547   0.894 0.371252
## burnout_frequency.C            -0.0098122  0.0065204  -1.505 0.132391
## burnout_frequency^4             0.0068047  0.0053359   1.275 0.202245
## remote_work_preference_1to5     0.0025447  0.0025439   1.000 0.317179
```

```
## countryAustralia                 1.7478878  0.0351511  49.725  < 2e-16 ***
## countryAustria                   1.4081097  0.0488114  28.848  < 2e-16 ***
## countryBangladesh               -0.5935144  0.0395474 -15.008  < 2e-16 ***
## countryBelgium                   1.3446026  0.0517286  25.993  < 2e-16 ***
## countryBrazil                    0.3613255  0.0322676  11.198  < 2e-16 ***
## countryCanada                    1.6659512  0.0321921  51.750  < 2e-16 ***
## countryChile                     0.4953260  0.0506667   9.776  < 2e-16 ***
## countryChina                     0.8369733  0.0368886  22.689  < 2e-16 ***
## countryColombia                  0.1579792  0.0398116   3.968 7.29e-05 ***
## countryCzech Republic            0.8879224  0.0409289  21.694  < 2e-16 ***
## countryDenmark                   1.4907098  0.0477627  31.211  < 2e-16 ***
## countryEgypt                    -0.1871600  0.0408019  -4.587 4.54e-06 ***
## countryFinland                   1.2934372  0.0477658  27.079  < 2e-16 ***
## countryFrance                    1.2956742  0.0332259  38.996  < 2e-16 ***
## countryGermany                   1.5615342  0.0313454  49.817  < 2e-16 ***
## countryGhana                    -0.4116155  0.0568308  -7.243 4.68e-13 ***
## countryIndia                     0.1784694  0.0300657   5.936 3.01e-09 ***
## countryIndonesia                -0.0941946  0.0373794  -2.520 0.011750 *
## countryIreland                   1.4778379  0.0450460  32.807  < 2e-16 ***
## countryIsrael                    1.6454724  0.0409482  40.184  < 2e-16 ***
## countryItaly                     1.0396279  0.0343976  30.224  < 2e-16 ***
## countryJapan                     1.2792078  0.0360952  35.440  < 2e-16 ***
## countryKenya                    -0.0862933  0.0448311  -1.925 0.054273 .
## countryMalaysia                  0.3011630  0.0542818   5.548 2.95e-08 ***
## countryMexico                    0.4508400  0.0411398  10.959  < 2e-16 ***
## countryNetherlands               1.4558390  0.0343028  42.441  < 2e-16 ***
## countryNew Zealand               1.4554718  0.0493988  29.464  < 2e-16 ***
## countryNigeria                  -0.1714358  0.0351672  -4.875 1.10e-06 ***
## countryNorway                    1.6292243  0.0457859  35.584  < 2e-16 ***
## countryPakistan                 -0.3959138  0.0365568 -10.830  < 2e-16 ***
## countryPeru                     -0.0052183  0.0594566  -0.088 0.930064
## countryPhilippines              -0.0942875  0.0416567  -2.263 0.023627 *
## countryPoland                    0.8473822  0.0331013  25.600  < 2e-16 ***
## countryPortugal                  0.7531051  0.0426099  17.674  < 2e-16 ***
## countryRomania                   0.7346063  0.0401259  18.308  < 2e-16 ***
## countryRussia                    0.4951252  0.0349649  14.161  < 2e-16 ***
## countrySingapore                 1.5849698  0.0465657  34.037  < 2e-16 ***
## countrySouth Africa              0.6137591  0.0441731  13.894  < 2e-16 ***
## countrySouth Korea               1.2087853  0.0366547  32.978  < 2e-16 ***
## countrySpain                     0.9600862  0.0354254  27.102  < 2e-16 ***
## countrySri Lanka                -0.2087545  0.0602195  -3.467 0.000529 ***
## countrySweden                    1.3737107  0.0393129  34.943  < 2e-16 ***
## countrySwitzerland               1.9873156  0.0401176  49.537  < 2e-16 ***
## countryThailand                  0.1457507  0.0568849   2.562 0.010414 *
## countryTurkey                    0.0144950  0.0353175   0.410 0.681505
```

```
## countryUkraine                    0.2930963  0.0368373   7.957 1.94e-15 ***
## countryUnited Kingdom             1.4950961  0.0315449  47.396  < 2e-16 ***
## countryUnited States              2.0766513  0.0298688  69.526  < 2e-16 ***
## countryVietnam                    0.0174778  0.0395538   0.442 0.658589
## open_source_contributor           0.0053387  0.0057791   0.924 0.355613
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2923 on 11342 degrees of freedom
## Multiple R-squared:  0.9225, Adjusted R-squared:  0.9218
## F-statistic:  1324 on 102 and 11342 DF,  p-value: < 2.2e-16
```

```r
AIC(model_full)
```

```
## [1] 4429.797
```

```r
BIC(model_full)
```

```
## [1] 5193.709
```

```r
drop1(model_full, test = "F")
```

Backward Elimination by p-value

```
## Single term deletions
##
## Model:
## log_salary ~ years_of_experience + age + coding_hours_per_week +
##     education_level + role + employment_status + company_size +
##     work_mode + gender + primary_os + ai_productivity_boost_pct +
##     ai_usage_frequency + burnout_frequency + remote_work_preference_1to5 +
##     country + open_source_contributor
##                            Df Sum of Sq    RSS      AIC   F value    Pr(>F)
## <none>                                   969.0 -28051.7
## years_of_experience         1     184.1 1153.1 -26063.0 2154.7626 < 2.2e-16
## age                         1       3.6  972.6 -28011.3   42.1147 8.966e-11
## coding_hours_per_week       1       0.5  969.6 -28047.7    6.0011   0.01431
## education_level             6      54.9 1023.9 -27433.2  107.0532 < 2.2e-16
## role                       15    1130.6 2099.7 -19232.2  882.2092 < 2.2e-16
## employment_status           4    1175.8 2144.8 -18966.5 3440.5048 < 2.2e-16
```

```
## company_size                  6       0.4  969.4 -28059.1    0.7561   0.60447
## work_mode                     2       0.2  969.2 -28053.3    1.1705   0.31026
## gender                        3       0.6  969.6 -28050.6    2.3469   0.07067
## primary_os                    2       0.1  969.1 -28054.5    0.6104   0.54314
## ai_productivity_boost_pct     1       0.0  969.1 -28053.2    0.5499   0.45839
## ai_usage_frequency            5       0.5  969.5 -28056.1    1.1046   0.35549
## burnout_frequency             4       0.5  969.5 -28053.7    1.4788   0.20564
## remote_work_preference_1to5   1       0.1  969.1 -28052.7    1.0006   0.31718
## country                      49    7123.2 8092.3  -3859.3 1701.4841 < 2.2e-16
## open_source_contributor       1       0.1  969.1 -28052.8    0.8534   0.35561
##
## <none>
## years_of_experience          ***
## age                          ***
## coding_hours_per_week        *
## education_level              ***
## role                         ***
## employment_status            ***
## company_size
## work_mode
## gender                       .
## primary_os
## ai_productivity_boost_pct
## ai_usage_frequency
## burnout_frequency
## remote_work_preference_1to5
## country                      ***
## open_source_contributor
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
model_backward_pval2 <- update(model_full, . ~ . - gender - primary_os - burnout_frequer
summary(model_backward_pval2)
```

```
##
## Call:
## lm(formula = log_salary ~ years_of_experience + age + coding_hours_per_week +
##     education_level + role + employment_status + company_size +
##     work_mode + ai_productivity_boost_pct + ai_usage_frequency +
##     remote_work_preference_1to5 + country + open_source_contributor,
##     data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -1.07334 -0.19477  0.00374  0.19908  1.57044
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   9.1227254  0.0393236 231.991  < 2e-16 ***
## years_of_experience           0.0364538  0.0007850  46.435  < 2e-16 ***
## age                           0.0051441  0.0007936   6.482 9.45e-11 ***
## coding_hours_per_week        -0.0008020  0.0003301  -2.429 0.015139 *
## education_level.L             0.3032472  0.0146130  20.752  < 2e-16 ***
## education_level.Q             0.0072403  0.0138698   0.522 0.601666
## education_level.C             0.0134392  0.0125167   1.074 0.282978
## education_level^4            -0.0148513  0.0106393  -1.396 0.162774
## education_level^5             0.0085677  0.0099897   0.858 0.391103
## education_level^6            -0.0158768  0.0086083  -1.844 0.065156 .
## roleCloud Engineer            0.0988848  0.0181581   5.446 5.27e-08 ***
## roleData Engineer             0.0525324  0.0151617   3.465 0.000533 ***
## roleData Scientist            0.0410931  0.0134766   3.049 0.002300 **
## roleDevOps / SRE              0.0943930  0.0132410   7.129 1.07e-12 ***
## roleEmbedded Developer        0.0015396  0.0215159   0.072 0.942958
## roleEngineering Manager       0.2000954  0.0178411  11.215  < 2e-16 ***
## roleFreelancer / Consultant   0.0007328  0.0153356   0.048 0.961888
## roleFront-End Developer      -0.1074954  0.0106085 -10.133  < 2e-16 ***
## roleFull-Stack Developer     -0.0494245  0.0091762  -5.386 7.34e-08 ***
## roleGame Developer           -0.1554806  0.0204132  -7.617 2.81e-14 ***
## roleMachine Learning Engineer 0.1397910  0.0140603   9.942  < 2e-16 ***
## roleMobile Developer         -0.0549237  0.0142036  -3.867 0.000111 ***
## roleQA / Test Engineer       -0.2355751  0.0176771 -13.327  < 2e-16 ***
## roleSecurity Engineer         0.1091724  0.0205713   5.307 1.14e-07 ***
## roleStudent / Intern         -1.3701443  0.0137977 -99.303  < 2e-16 ***
## employment_statusFull-time    0.0074720  0.0100394   0.744 0.456734
## employment_statusPart-time    0.0048766  0.0152631   0.320 0.749350
## employment_statusStudent     -1.5867507  0.0185561 -85.511  < 2e-16 ***
## employment_statusUnemployed  -1.5988170  0.0235175 -67.984  < 2e-16 ***
## company_size.L                0.0003079  0.0079821   0.039 0.969231
## company_size.Q                0.0095592  0.0073241   1.305 0.191866
## company_size.C               -0.0035329  0.0077497  -0.456 0.648488
## company_size^4                0.0036948  0.0079651   0.464 0.642742
## company_size^5                0.0099540  0.0075188   1.324 0.185566
## company_size^6               -0.0024994  0.0067015  -0.373 0.709184
## work_modeIn-Office            0.0026338  0.0073426   0.359 0.719826
## work_modeRemote              -0.0073954  0.0061877  -1.195 0.232041
## ai_productivity_boost_pct    -0.0001694  0.0002245  -0.754 0.450611
## ai_usage_frequency.L          0.0104018  0.0098905   1.052 0.292964
## ai_usage_frequency.Q         -0.0056759  0.0075279  -0.754 0.450870
## ai_usage_frequency.C         -0.0052726  0.0075185  -0.701 0.483137
```

```
## ai_usage_frequency^4          0.0147334  0.0074523   1.977 0.048062 *
## ai_usage_frequency^5         -0.0087474  0.0076964  -1.137 0.255746
## remote_work_preference_1to5   0.0024476  0.0025435   0.962 0.335932
## countryAustralia              1.7482398  0.0351245  49.773  < 2e-16 ***
## countryAustria                1.4079599  0.0487996  28.852  < 2e-16 ***
## countryBangladesh            -0.5936503  0.0395424 -15.013  < 2e-16 ***
## countryBelgium                1.3448470  0.0517080  26.009  < 2e-16 ***
## countryBrazil                 0.3624533  0.0322529  11.238  < 2e-16 ***
## countryCanada                 1.6663840  0.0321752  51.791  < 2e-16 ***
## countryChile                  0.4967149  0.0506467   9.807  < 2e-16 ***
## countryChina                  0.8371706  0.0368678  22.707  < 2e-16 ***
## countryColombia               0.1583233  0.0397860   3.979 6.95e-05 ***
## countryCzech Republic         0.8882067  0.0409069  21.713  < 2e-16 ***
## countryDenmark                1.4900926  0.0477507  31.206  < 2e-16 ***
## countryEgypt                 -0.1892111  0.0408004  -4.637 3.57e-06 ***
## countryFinland                1.2916153  0.0477444  27.053  < 2e-16 ***
## countryFrance                 1.2960615  0.0332151  39.020  < 2e-16 ***
## countryGermany                1.5624443  0.0313250  49.878  < 2e-16 ***
## countryGhana                 -0.4095468  0.0568131  -7.209 6.01e-13 ***
## countryIndia                  0.1789082  0.0300477   5.954 2.69e-09 ***
## countryIndonesia             -0.0925566  0.0373620  -2.477 0.013253 *
## countryIreland                1.4769086  0.0450256  32.802  < 2e-16 ***
## countryIsrael                 1.6460677  0.0409133  40.233  < 2e-16 ***
## countryItaly                  1.0396535  0.0343833  30.237  < 2e-16 ***
## countryJapan                  1.2801047  0.0360900  35.470  < 2e-16 ***
## countryKenya                 -0.0859558  0.0448160  -1.918 0.055139 .
## countryMalaysia               0.3029247  0.0542643   5.582 2.43e-08 ***
## countryMexico                 0.4509926  0.0411178  10.968  < 2e-16 ***
## countryNetherlands            1.4566052  0.0342877  42.482  < 2e-16 ***
## countryNew Zealand            1.4553794  0.0493799  29.473  < 2e-16 ***
## countryNigeria               -0.1721637  0.0351539  -4.897 9.84e-07 ***
## countryNorway                 1.6281843  0.0457793  35.566  < 2e-16 ***
## countryPakistan              -0.3973620  0.0365404 -10.875  < 2e-16 ***
## countryPeru                  -0.0027364  0.0594431  -0.046 0.963284
## countryPhilippines           -0.0942847  0.0416429  -2.264 0.023585 *
## countryPoland                 0.8480736  0.0330945  25.626  < 2e-16 ***
## countryPortugal               0.7547876  0.0425823  17.725  < 2e-16 ***
## countryRomania                0.7362429  0.0401070  18.357  < 2e-16 ***
## countryRussia                 0.4945776  0.0349532  14.150  < 2e-16 ***
## countrySingapore              1.5836961  0.0465577  34.016  < 2e-16 ***
## countrySouth Africa           0.6162595  0.0441513  13.958  < 2e-16 ***
## countrySouth Korea            1.2101445  0.0366440  33.024  < 2e-16 ***
## countrySpain                  0.9610135  0.0354064  27.142  < 2e-16 ***
## countrySri Lanka             -0.2106844  0.0602117  -3.499 0.000469 ***
## countrySweden                 1.3729560  0.0392842  34.949  < 2e-16 ***
```

```
## countrySwitzerland           1.9885617  0.0401066  49.582  < 2e-16 ***
## countryThailand              0.1465797  0.0568548   2.578 0.009946 **
## countryTurkey                0.0145740  0.0352986   0.413 0.679705
## countryUkraine               0.2918829  0.0368225   7.927 2.46e-15 ***
## countryUnited Kingdom        1.4952991  0.0315361  47.415  < 2e-16 ***
## countryUnited States         2.0773014  0.0298522  69.586  < 2e-16 ***
## countryVietnam               0.0185582  0.0395436   0.469 0.638857
## open_source_contributor      0.0053604  0.0057789   0.928 0.353638
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2924 on 11351 degrees of freedom
## Multiple R-squared:  0.9224, Adjusted R-squared:  0.9218
## F-statistic:  1451 on 93 and 11351 DF,  p-value: < 2.2e-16
```

```r
drop1(model_backward_pval2, test = "F")
```

```
## Single term deletions
##
## Model:
## log_salary ~ years_of_experience + age + coding_hours_per_week +
##     education_level + role + employment_status + company_size +
##     work_mode + ai_productivity_boost_pct + ai_usage_frequency +
##     remote_work_preference_1to5 + country + open_source_contributor
##                             Df Sum of Sq    RSS      AIC   F value    Pr(>F)
## <none>                                    970.2 -28055.6
## years_of_experience          1     184.3 1154.5 -26067.1 2156.2172 < 2.2e-16
## age                          1       3.6  973.8 -28015.3   42.0115  9.45e-11
## coding_hours_per_week        1       0.5  970.7 -28051.6    5.9020   0.01514
## education_level              6      55.2 1025.4 -27434.3  107.6295 < 2.2e-16
## role                        15    1131.3 2101.5 -19240.0  882.3396 < 2.2e-16
## employment_status            4    1177.1 2147.3 -18971.2 3442.7731 < 2.2e-16
## company_size                 6       0.4  970.6 -28063.0    0.7578   0.60312
## work_mode                    2       0.2  970.4 -28057.3    1.1510   0.31637
## ai_productivity_boost_pct    1       0.0  970.3 -28057.0    0.5692   0.45061
## ai_usage_frequency           5       0.5  970.7 -28059.9    1.1337   0.33992
## remote_work_preference_1to5  1       0.1  970.3 -28056.6    0.9260   0.33593
## country                     49    7134.8 8105.0  -3859.3 1703.4945 < 2.2e-16
## open_source_contributor      1       0.1  970.3 -28056.7    0.8604   0.35364
##
## <none>
## years_of_experience         ***
## age                         ***
## coding_hours_per_week       *
```

```
## education_level            ***
## role                       ***
## employment_status          ***
## company_size
## work_mode
## ai_productivity_boost_pct
## ai_usage_frequency
## remote_work_preference_1to5
## country                    ***
## open_source_contributor
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
model_step_aic <- stepAIC(model_full, direction = "both", trace = FALSE)
summary(model_step_aic)
```

Stepwise AIC

```
##
## Call:
## lm(formula = log_salary ~ years_of_experience + age + coding_hours_per_week +
##     education_level + role + employment_status + gender + country,
##     data = data_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.0721 -0.1953  0.0049  0.1995  1.5813
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  9.1319846  0.0374999 243.520  < 2e-16 ***
## years_of_experience          0.0364809  0.0007845  46.504  < 2e-16 ***
## age                          0.0051049  0.0007931   6.437 1.27e-10 ***
## coding_hours_per_week       -0.0008017  0.0003299  -2.430 0.015094 *
## education_level.L            0.3025503  0.0146055  20.715  < 2e-16 ***
## education_level.Q            0.0070215  0.0138599   0.507 0.612442
## education_level.C            0.0136356  0.0125102   1.090 0.275755
## education_level^4           -0.0152186  0.0106307  -1.432 0.152294
## education_level^5            0.0078318  0.0099823   0.785 0.432724
## education_level^6           -0.0154189  0.0086027  -1.792 0.073108 .
## roleCloud Engineer           0.0993975  0.0181505   5.476 4.44e-08 ***
## roleData Engineer            0.0526812  0.0151484   3.478 0.000508 ***
```

```
## roleData Scientist              0.0418878  0.0134610   3.112 0.001864 **
## roleDevOps / SRE                0.0951049  0.0132307   7.188 6.98e-13 ***
## roleEmbedded Developer          0.0017095  0.0215017   0.080 0.936632
## roleEngineering Manager         0.2000554  0.0178222  11.225  < 2e-16 ***
## roleFreelancer / Consultant     0.0014292  0.0153270   0.093 0.925710
## roleFront-End Developer        -0.1071623  0.0105984 -10.111  < 2e-16 ***
## roleFull-Stack Developer       -0.0490777  0.0091671  -5.354 8.78e-08 ***
## roleGame Developer             -0.1553356  0.0203976  -7.615 2.84e-14 ***
## roleMachine Learning Engineer   0.1400940  0.0140446   9.975  < 2e-16 ***
## roleMobile Developer           -0.0545494  0.0141933  -3.843 0.000122 ***
## roleQA / Test Engineer         -0.2336041  0.0176707 -13.220  < 2e-16 ***
## roleSecurity Engineer           0.1101032  0.0205588   5.356 8.70e-08 ***
## roleStudent / Intern           -1.3689458  0.0137851 -99.306  < 2e-16 ***
## employment_statusFull-time      0.0072718  0.0100310   0.725 0.468508
## employment_statusPart-time      0.0052134  0.0152487   0.342 0.732439
## employment_statusStudent       -1.5868063  0.0185350 -85.612  < 2e-16 ***
## employment_statusUnemployed    -1.5993659  0.0235037 -68.047  < 2e-16 ***
## genderNon-binary               -0.0171192  0.0202189  -0.847 0.397185
## genderPrefer not to say         0.0428119  0.0193445   2.213 0.026909 *
## genderWoman                    -0.0106913  0.0098639  -1.084 0.278442
## countryAustralia                1.7461723  0.0350959  49.754  < 2e-16 ***
## countryAustria                  1.4038702  0.0487471  28.799  < 2e-16 ***
## countryBangladesh              -0.5938531  0.0395177 -15.028  < 2e-16 ***
## countryBelgium                  1.3411798  0.0516733  25.955  < 2e-16 ***
## countryBrazil                   0.3602975  0.0322298  11.179  < 2e-16 ***
## countryCanada                   1.6636879  0.0321546  51.740  < 2e-16 ***
## countryChile                    0.4962957  0.0506095   9.806  < 2e-16 ***
## countryChina                    0.8349159  0.0368390  22.664  < 2e-16 ***
## countryColombia                 0.1564674  0.0397584   3.935 8.35e-05 ***
## countryCzech Republic           0.8870539  0.0408857  21.696  < 2e-16 ***
## countryDenmark                  1.4885894  0.0477219  31.193  < 2e-16 ***
## countryEgypt                   -0.1892337  0.0407746  -4.641 3.51e-06 ***
## countryFinland                  1.2893997  0.0477144  27.023  < 2e-16 ***
## countryFrance                   1.2942995  0.0331889  38.998  < 2e-16 ***
## countryGermany                  1.5608333  0.0313049  49.859  < 2e-16 ***
## countryGhana                   -0.4122973  0.0567854  -7.261 4.10e-13 ***
## countryIndia                    0.1771376  0.0300206   5.901 3.73e-09 ***
## countryIndonesia               -0.0944726  0.0373281  -2.531 0.011391 *
## countryIreland                  1.4775113  0.0449796  32.848  < 2e-16 ***
## countryIsrael                   1.6447075  0.0408883  40.224  < 2e-16 ***
## countryItaly                    1.0388074  0.0343530  30.239  < 2e-16 ***
## countryJapan                    1.2775253  0.0360590  35.429  < 2e-16 ***
## countryKenya                   -0.0879588  0.0447921  -1.964 0.049588 *
## countryMalaysia                 0.3021149  0.0542138   5.573 2.57e-08 ***
## countryMexico                   0.4497487  0.0410882  10.946  < 2e-16 ***
```

```
## countryNetherlands              1.4550535  0.0342632  42.467  < 2e-16 ***
## countryNew Zealand              1.4542660  0.0493247  29.484  < 2e-16 ***
## countryNigeria                 -0.1731126  0.0351338  -4.927 8.46e-07 ***
## countryNorway                   1.6268187  0.0457371  35.569  < 2e-16 ***
## countryPakistan                -0.3986154  0.0365192 -10.915  < 2e-16 ***
## countryPeru                    -0.0082198  0.0593802  -0.138 0.889906
## countryPhilippines             -0.0955830  0.0416100  -2.297 0.021630 *
## countryPoland                   0.8447562  0.0330622  25.551  < 2e-16 ***
## countryPortugal                 0.7524953  0.0425542  17.683  < 2e-16 ***
## countryRomania                  0.7336612  0.0400696  18.310  < 2e-16 ***
## countryRussia                   0.4934532  0.0349283  14.128  < 2e-16 ***
## countrySingapore                1.5831731  0.0465122  34.038  < 2e-16 ***
## countrySouth Africa             0.6120327  0.0441106  13.875  < 2e-16 ***
## countrySouth Korea              1.2065933  0.0366182  32.951  < 2e-16 ***
## countrySpain                    0.9597044  0.0353745  27.130  < 2e-16 ***
## countrySri Lanka               -0.2122421  0.0601471  -3.529 0.000419 ***
## countrySweden                   1.3717648  0.0392505  34.949  < 2e-16 ***
## countrySwitzerland              1.9861255  0.0400835  49.550  < 2e-16 ***
## countryThailand                 0.1449399  0.0568029   2.552 0.010735 *
## countryTurkey                   0.0152047  0.0352640   0.431 0.666355
## countryUkraine                  0.2918585  0.0367958   7.932 2.36e-15 ***
## countryUnited Kingdom           1.4938656  0.0315127  47.405  < 2e-16 ***
## countryUnited States            2.0754532  0.0298268  69.584  < 2e-16 ***
## countryVietnam                  0.0158159  0.0395050   0.400 0.688905
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2923 on 11364 degrees of freedom
## Multiple R-squared:  0.9224, Adjusted R-squared:  0.9218
## F-statistic:  1688 on 80 and 11364 DF,  p-value: < 2.2e-16
```

```r
AIC(model_step_aic)
```

```
## [1] 4407.182
```

```r
BIC(model_step_aic)
```

```
## [1] 5009.497
```

```r
n <- nrow(data_clean)
model_step_bic <- stepAIC(model_full, direction = "both", k = log(n), trace = FALSE)
summary(model_step_bic)
```

Stepwise BIC

```
##
## Call:
## lm(formula = log_salary ~ years_of_experience + age + education_level +
##     role + employment_status + country, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.08131 -0.19451  0.00548  0.20034  1.58556
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   9.1097679  0.0364526 249.907  < 2e-16 ***
## years_of_experience           0.0364757  0.0007847  46.484  < 2e-16 ***
## age                           0.0051010  0.0007933   6.430 1.33e-10 ***
## education_level.L             0.3025459  0.0146074  20.712  < 2e-16 ***
## education_level.Q             0.0075329  0.0138641   0.543 0.586907
## education_level.C             0.0134339  0.0125141   1.073 0.283070
## education_level^4            -0.0152650  0.0106347  -1.435 0.151204
## education_level^5             0.0083642  0.0099845   0.838 0.402209
## education_level^6            -0.0152901  0.0086050  -1.777 0.075612 .
## roleCloud Engineer            0.0986565  0.0181517   5.435 5.59e-08 ***
## roleData Engineer             0.0524027  0.0151525   3.458 0.000546 ***
## roleData Scientist            0.0415544  0.0134641   3.086 0.002031 **
## roleDevOps / SRE              0.0945115  0.0132343   7.141 9.80e-13 ***
## roleEmbedded Developer        0.0002187  0.0215037   0.010 0.991885
## roleEngineering Manager       0.2003955  0.0178274  11.241  < 2e-16 ***
## roleFreelancer / Consultant   0.0009762  0.0153277   0.064 0.949220
## roleFront-End Developer      -0.1076613  0.0106014 -10.155  < 2e-16 ***
## roleFull-Stack Developer     -0.0494451  0.0091695  -5.392 7.09e-08 ***
## roleGame Developer           -0.1563014  0.0204027  -7.661 2.00e-14 ***
## roleMachine Learning Engineer  0.1397598  0.0140478   9.949  < 2e-16 ***
## roleMobile Developer         -0.0556264  0.0141951  -3.919 8.95e-05 ***
## roleQA / Test Engineer       -0.2352704  0.0176708 -13.314  < 2e-16 ***
## roleSecurity Engineer         0.1099138  0.0205592   5.346 9.16e-08 ***
## roleStudent / Intern         -1.3699081  0.0137876 -99.358  < 2e-16 ***
## employment_statusFull-time    0.0014429  0.0097246   0.148 0.882046
## employment_statusPart-time    0.0046148  0.0152514   0.303 0.762212
## employment_statusStudent     -1.5806779  0.0183738 -86.029  < 2e-16 ***
## employment_statusUnemployed  -1.5987564  0.0235098 -68.004  < 2e-16 ***
## countryAustralia              1.7464708  0.0351049  49.750  < 2e-16 ***
## countryAustria                1.4036884  0.0487632  28.786  < 2e-16 ***
## countryBangladesh            -0.5931628  0.0395258 -15.007  < 2e-16 ***
## countryBelgium                1.3410930  0.0516868  25.947  < 2e-16 ***
```

```
## countryBrazil                  0.3616941  0.0322378  11.220  < 2e-16 ***
## countryCanada                  1.6649340  0.0321624  51.767  < 2e-16 ***
## countryChile                   0.4973552  0.0506265   9.824  < 2e-16 ***
## countryChina                   0.8353512  0.0368522  22.668  < 2e-16 ***
## countryColombia                0.1577787  0.0397666   3.968 7.30e-05 ***
## countryCzech Republic          0.8855680  0.0408971  21.654  < 2e-16 ***
## countryDenmark                 1.4898849  0.0477352  31.211  < 2e-16 ***
## countryEgypt                  -0.1900784  0.0407894  -4.660 3.20e-06 ***
## countryFinland                 1.2892715  0.0477242  27.015  < 2e-16 ***
## countryFrance                  1.2949138  0.0331997  39.004  < 2e-16 ***
## countryGermany                 1.5611476  0.0313133  49.856  < 2e-16 ***
## countryGhana                  -0.4109288  0.0567976  -7.235 4.96e-13 ***
## countryIndia                   0.1778427  0.0300309   5.922 3.27e-09 ***
## countryIndonesia              -0.0933300  0.0373406  -2.499 0.012454 *
## countryIreland                 1.4791016  0.0449893  32.877  < 2e-16 ***
## countryIsrael                  1.6468565  0.0408959  40.269  < 2e-16 ***
## countryItaly                   1.0389226  0.0343652  30.232  < 2e-16 ***
## countryJapan                   1.2790434  0.0360704  35.460  < 2e-16 ***
## countryKenya                  -0.0863146  0.0448031  -1.927 0.054063 .
## countryMalaysia                0.3022109  0.0542310   5.573 2.57e-08 ***
## countryMexico                  0.4510160  0.0411019  10.973  < 2e-16 ***
## countryNetherlands             1.4554659  0.0342745  42.465  < 2e-16 ***
## countryNew Zealand             1.4535393  0.0493424  29.458  < 2e-16 ***
## countryNigeria                -0.1718897  0.0351368  -4.892 1.01e-06 ***
## countryNorway                  1.6260965  0.0457514  35.542  < 2e-16 ***
## countryPakistan               -0.3991766  0.0365317 -10.927  < 2e-16 ***
## countryPeru                   -0.0078608  0.0593972  -0.132 0.894715
## countryPhilippines            -0.0957533  0.0416176  -2.301 0.021422 *
## countryPoland                  0.8459060  0.0330727  25.577  < 2e-16 ***
## countryPortugal                0.7547321  0.0425618  17.733  < 2e-16 ***
## countryRomania                 0.7343241  0.0400815  18.321  < 2e-16 ***
## countryRussia                  0.4937921  0.0349393  14.133  < 2e-16 ***
## countrySingapore               1.5842966  0.0465228  34.054  < 2e-16 ***
## countrySouth Africa            0.6144611  0.0441188  13.927  < 2e-16 ***
## countrySouth Korea             1.2081663  0.0366287  32.984  < 2e-16 ***
## countrySpain                   0.9601681  0.0353855  27.135  < 2e-16 ***
## countrySri Lanka              -0.2137669  0.0601630  -3.553 0.000382 ***
## countrySweden                  1.3736087  0.0392604  34.987  < 2e-16 ***
## countrySwitzerland             1.9884056  0.0400904  49.598  < 2e-16 ***
## countryThailand                0.1451798  0.0568213   2.555 0.010631 *
## countryTurkey                  0.0148525  0.0352755   0.421 0.673731
## countryUkraine                 0.2916903  0.0368071   7.925 2.50e-15 ***
## countryUnited Kingdom          1.4946275  0.0315235  47.413  < 2e-16 ***
## countryUnited States           2.0762105  0.0298364  69.587  < 2e-16 ***
## countryVietnam                 0.0156733  0.0395133   0.397 0.691626
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2924 on 11368 degrees of freedom
## Multiple R-squared:  0.9223, Adjusted R-squared:  0.9218
## F-statistic:  1775 on 76 and 11368 DF,  p-value: < 2.2e-16
```

```r
AIC(model_step_bic)
```

```
## [1] 4412.148
```

```r
BIC(model_step_bic)
```

```
## [1] 4985.082
```

```r
model_theory <- lm(
  log_salary ~ years_of_experience + education_level + role +
    employment_status + company_size + work_mode + country,
  data = data_clean
)

summary(model_theory)
```

Theory-Driven Reduced Model

```
##
## Call:
## lm(formula = log_salary ~ years_of_experience + education_level +
##     role + employment_status + company_size + work_mode + country,
##     data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.08397 -0.19517  0.00564  0.20016  1.56985
##
## Coefficients:
##                                 Estimate Std. Error  t value Pr(>|t|)
## (Intercept)                    9.2299572  0.0317373  290.824  < 2e-16 ***
## years_of_experience            0.0409464  0.0003658  111.933  < 2e-16 ***
## education_level.L              0.3028150  0.0146384   20.686  < 2e-16 ***
```

```
## education_level.Q              0.0072685  0.0138918    0.523 0.600831
## education_level.C              0.0148768  0.0125377    1.187 0.235425
## education_level^4             -0.0153239  0.0106587   -1.438 0.150552
## education_level^5              0.0089350  0.0100071    0.893 0.371947
## education_level^6             -0.0155882  0.0086236   -1.808 0.070692 .
## roleCloud Engineer             0.0986390  0.0181889    5.423 5.98e-08 ***
## roleData Engineer              0.0534685  0.0151831    3.522 0.000431 ***
## roleData Scientist             0.0420390  0.0134946    3.115 0.001843 **
## roleDevOps / SRE               0.0946140  0.0132638    7.133 1.04e-12 ***
## roleEmbedded Developer        -0.0004387  0.0215511   -0.020 0.983761
## roleEngineering Manager        0.2078425  0.0178327   11.655  < 2e-16 ***
## roleFreelancer / Consultant    0.0010311  0.0153608    0.067 0.946483
## roleFront-End Developer       -0.1069729  0.0106237  -10.069  < 2e-16 ***
## roleFull-Stack Developer      -0.0493921  0.0091904   -5.374 7.84e-08 ***
## roleGame Developer            -0.1565612  0.0204474   -7.657 2.06e-14 ***
## roleMachine Learning Engineer  0.1410825  0.0140749   10.024  < 2e-16 ***
## roleMobile Developer          -0.0556588  0.0142268   -3.912 9.20e-05 ***
## roleQA / Test Engineer        -0.2355224  0.0177068  -13.301  < 2e-16 ***
## roleSecurity Engineer          0.1095534  0.0206020    5.318 1.07e-07 ***
## roleStudent / Intern          -1.3842615  0.0136402 -101.484  < 2e-16 ***
## employment_statusFull-time     0.0002210  0.0097438    0.023 0.981903
## employment_statusPart-time     0.0020247  0.0152762    0.133 0.894561
## employment_statusStudent      -1.5808375  0.0184186  -85.828  < 2e-16 ***
## employment_statusUnemployed   -1.5999181  0.0235562  -67.919  < 2e-16 ***
## company_size.L                 0.0002295  0.0079917    0.029 0.977091
## company_size.Q                 0.0086606  0.0073358    1.181 0.237786
## company_size.C                -0.0029129  0.0077630   -0.375 0.707498
## company_size^4                 0.0032744  0.0079789    0.410 0.681532
## company_size^5                 0.0106513  0.0075327    1.414 0.157386
## company_size^6                -0.0025759  0.0067115   -0.384 0.701133
## work_modeIn-Office             0.0032463  0.0073561    0.441 0.658995
## work_modeRemote               -0.0066553  0.0061977   -1.074 0.282915
## countryAustralia               1.7488148  0.0351829   49.706  < 2e-16 ***
## countryAustria                 1.4063325  0.0488666   28.779  < 2e-16 ***
## countryBangladesh             -0.5923692  0.0396030  -14.958  < 2e-16 ***
## countryBelgium                 1.3442076  0.0517939   25.953  < 2e-16 ***
## countryBrazil                  0.3635492  0.0323000   11.255  < 2e-16 ***
## countryCanada                  1.6661859  0.0322277   51.700  < 2e-16 ***
## countryChile                   0.4974678  0.0507287    9.806  < 2e-16 ***
## countryChina                   0.8368916  0.0369347   22.659  < 2e-16 ***
## countryColombia                0.1599788  0.0398462    4.015 5.99e-05 ***
## countryCzech Republic          0.8870903  0.0409819   21.646  < 2e-16 ***
## countryDenmark                 1.4865958  0.0478258   31.084  < 2e-16 ***
## countryEgypt                  -0.1876278  0.0408664   -4.591 4.45e-06 ***
## countryFinland                 1.2896955  0.0478239   26.968  < 2e-16 ***
```

```
## countryFrance                  1.2963037  0.0332665   38.967  < 2e-16 ***
## countryGermany                 1.5628613  0.0313762   49.810  < 2e-16 ***
## countryGhana                  -0.4144973  0.0569138   -7.283 3.48e-13 ***
## countryIndia                   0.1795384  0.0300940    5.966 2.51e-09 ***
## countryIndonesia              -0.0891519  0.0374144   -2.383 0.017197 *
## countryIreland                 1.4828872  0.0450767   32.897  < 2e-16 ***
## countryIsrael                  1.6481859  0.0409766   40.223  < 2e-16 ***
## countryItaly                   1.0398928  0.0344460   30.189  < 2e-16 ***
## countryJapan                   1.2812723  0.0361426   35.450  < 2e-16 ***
## countryKenya                  -0.0843139  0.0448912   -1.878 0.060382 .
## countryMalaysia                0.3026668  0.0543464    5.569 2.62e-08 ***
## countryMexico                  0.4548945  0.0411890   11.044  < 2e-16 ***
## countryNetherlands             1.4579634  0.0343428   42.453  < 2e-16 ***
## countryNew Zealand             1.4547575  0.0494656   29.409  < 2e-16 ***
## countryNigeria                -0.1721144  0.0352091   -4.888 1.03e-06 ***
## countryNorway                  1.6300676  0.0458414   35.559  < 2e-16 ***
## countryPakistan               -0.3962819  0.0366063  -10.826  < 2e-16 ***
## countryPeru                   -0.0062348  0.0595318   -0.105 0.916592
## countryPhilippines            -0.0986185  0.0417086   -2.364 0.018073 *
## countryPoland                  0.8470235  0.0331423   25.557  < 2e-16 ***
## countryPortugal                0.7558411  0.0426484   17.723  < 2e-16 ***
## countryRomania                 0.7336991  0.0401638   18.268  < 2e-16 ***
## countryRussia                  0.4964952  0.0350052   14.183  < 2e-16 ***
## countrySingapore               1.5835507  0.0466195   33.968  < 2e-16 ***
## countrySouth Africa            0.6203403  0.0442165   14.030  < 2e-16 ***
## countrySouth Korea             1.2085026  0.0367035   32.926  < 2e-16 ***
## countrySpain                   0.9607526  0.0354683   27.088  < 2e-16 ***
## countrySri Lanka              -0.2101186  0.0602809   -3.486 0.000493 ***
## countrySweden                  1.3787519  0.0393416   35.046  < 2e-16 ***
## countrySwitzerland             1.9880325  0.0401753   49.484  < 2e-16 ***
## countryThailand                0.1483162  0.0569312    2.605 0.009194 **
## countryTurkey                  0.0139866  0.0353567    0.396 0.692420
## countryUkraine                 0.2912227  0.0368808    7.896 3.14e-15 ***
## countryUnited Kingdom          1.4968757  0.0315886   47.387  < 2e-16 ***
## countryUnited States           2.0775541  0.0298995   69.484  < 2e-16 ***
## countryVietnam                 0.0187741  0.0396032    0.474 0.635470
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2929 on 11361 degrees of freedom
## Multiple R-squared:  0.922,  Adjusted R-squared:  0.9215
## F-statistic:  1619 on 83 and 11361 DF,  p-value: < 2.2e-16
```

```r
AIC(model_theory)
```

```
## [1] 4461.216
```

```r
BIC(model_theory)
```

```
## [1] 5085.567
```

```r
vif(model_full)
```

VIF Check — Multicollinearity

```
##                                GVIF Df GVIF^(1/(2*Df))
## years_of_experience        5.271419  1        2.295957
## age                        5.456459  1        2.335907
## coding_hours_per_week      1.191233  1        1.091436
## education_level            1.053931  6        1.004387
## role                       1.322811 15        1.009369
## employment_status          1.225800  4        1.025776
## company_size               1.050559  6        1.004119
## work_mode                  1.015871  2        1.003944
## gender                     1.024460  3        1.004036
## primary_os                 1.018933  2        1.004700
## ai_productivity_boost_pct  1.970907  1        1.403890
## ai_usage_frequency         2.028147  5        1.073272
## burnout_frequency          1.036069  4        1.004439
## remote_work_preference_1to5 1.008348  1        1.004165
## country                    1.257315 49        1.002339
## open_source_contributor    1.010686  1        1.005329
```

```r
model_no_collinear <- lm(
  log_salary ~ years_of_experience + coding_hours_per_week +
    education_level + role + employment_status + company_size +
    work_mode + gender + primary_os + ai_productivity_boost_pct +
    ai_usage_frequency + burnout_frequency + remote_work_preference_1to5 +
    country + open_source_contributor,
  data = data_clean
)

vif(model_no_collinear)
```

```
##                                GVIF Df GVIF^(1/(2*Df))
## years_of_experience        1.141013  1        1.068182
## coding_hours_per_week      1.191170  1        1.091407
## education_level            1.053469  6        1.004350
## role                       1.272647 15        1.008069
## employment_status          1.224602  4        1.025650
## company_size               1.050082  6        1.004081
## work_mode                  1.015673  2        1.003895
## gender                     1.024375  3        1.004022
## primary_os                 1.018788  2        1.004664
## ai_productivity_boost_pct   1.970792  1        1.403849
## ai_usage_frequency         2.027795  5        1.073254
## burnout_frequency          1.035925  4        1.004422
## remote_work_preference_1to5 1.008344  1        1.004163
## country                    1.251947 49        1.002295
## open_source_contributor    1.010685  1        1.005328
```

```r
summary(model_no_collinear)
```

```
##
## Call:
## lm(formula = log_salary ~ years_of_experience + coding_hours_per_week +
##     education_level + role + employment_status + company_size +
##     work_mode + gender + primary_os + ai_productivity_boost_pct +
##     ai_usage_frequency + burnout_frequency + remote_work_preference_1to5 +
##     country + open_source_contributor, data = data_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.0861 -0.1948  0.0054  0.1995  1.5680
##
## Coefficients:
##                                Estimate Std. Error  t value Pr(>|t|)
## (Intercept)                   9.240e+00  3.520e-02  262.508  < 2e-16 ***
## years_of_experience           4.095e-02  3.659e-04  111.919  < 2e-16 ***
## coding_hours_per_week        -7.931e-04  3.308e-04   -2.398 0.016504 *
## education_level.L             3.026e-01  1.465e-02   20.654  < 2e-16 ***
## education_level.Q             6.255e-03  1.390e-02    0.450 0.652686
## education_level.C             1.483e-02  1.254e-02    1.183 0.236913
## education_level^4            -1.522e-02  1.066e-02   -1.428 0.153451
## education_level^5             8.089e-03  1.001e-02    0.808 0.419059
## education_level^6            -1.603e-02  8.625e-03   -1.859 0.063110 .
## roleCloud Engineer            9.962e-02  1.819e-02    5.475 4.46e-08 ***
## roleData Engineer             5.362e-02  1.519e-02    3.530 0.000417 ***
```

```
## roleData Scientist             4.171e-02  1.350e-02    3.089 0.002012 **
## roleDevOps / SRE               9.491e-02  1.327e-02    7.153 9.03e-13 ***
## roleEmbedded Developer         1.140e-03  2.156e-02    0.053 0.957841
## roleEngineering Manager        2.071e-01  1.784e-02   11.611  < 2e-16 ***
## roleFreelancer / Consultant    9.834e-04  1.537e-02    0.064 0.948973
## roleFront-End Developer       -1.068e-01  1.063e-02  -10.051  < 2e-16 ***
## roleFull-Stack Developer      -4.951e-02  9.193e-03   -5.386 7.34e-08 ***
## roleGame Developer            -1.551e-01  2.045e-02   -7.582 3.65e-14 ***
## roleMachine Learning Engineer  1.410e-01  1.409e-02   10.009  < 2e-16 ***
## roleMobile Developer          -5.465e-02  1.423e-02   -3.840 0.000124 ***
## roleQA / Test Engineer        -2.354e-01  1.772e-02  -13.290  < 2e-16 ***
## roleSecurity Engineer          1.092e-01  2.062e-02    5.299 1.19e-07 ***
## roleStudent / Intern          -1.384e+00  1.365e-02 -101.419  < 2e-16 ***
## employment_statusFull-time     6.196e-03  1.006e-02    0.616 0.537826
## employment_statusPart-time     2.305e-03  1.529e-02    0.151 0.880192
## employment_statusStudent      -1.586e+00  1.859e-02  -85.279  < 2e-16 ***
## employment_statusUnemployed   -1.600e+00  2.356e-02  -67.931  < 2e-16 ***
## company_size.L                 8.825e-04  7.997e-03    0.110 0.912131
## company_size.Q                 9.127e-03  7.338e-03    1.244 0.213610
## company_size.C                -3.137e-03  7.763e-03   -0.404 0.686120
## company_size^4                 3.656e-03  7.981e-03    0.458 0.646926
## company_size^5                 1.006e-02  7.534e-03    1.336 0.181664
## company_size^6                -2.368e-03  6.714e-03   -0.353 0.724338
## work_modeIn-Office             2.830e-03  7.358e-03    0.385 0.700512
## work_modeRemote               -6.950e-03  6.200e-03   -1.121 0.262285
## genderNon-binary              -1.742e-02  2.026e-02   -0.859 0.390096
## genderPrefer not to say        4.361e-02  1.939e-02    2.249 0.024503 *
## genderWoman                   -1.054e-02  9.893e-03   -1.065 0.286690
## primary_osmacOS                6.223e-03  7.238e-03    0.860 0.389984
## primary_osWindows              1.649e-05  6.876e-03    0.002 0.998087
## ai_productivity_boost_pct     -1.554e-04  2.249e-04   -0.691 0.489729
## ai_usage_frequency.L           1.035e-02  9.910e-03    1.044 0.296336
## ai_usage_frequency.Q          -4.782e-03  7.544e-03   -0.634 0.526231
## ai_usage_frequency.C          -5.418e-03  7.532e-03   -0.719 0.471974
## ai_usage_frequency^4           1.459e-02  7.467e-03    1.954 0.050755 .
## ai_usage_frequency^5          -8.595e-03  7.711e-03   -1.115 0.265056
## burnout_frequency.L            6.772e-03  8.454e-03    0.801 0.423105
## burnout_frequency.Q            7.091e-03  7.568e-03    0.937 0.348780
## burnout_frequency.C           -9.571e-03  6.532e-03   -1.465 0.142900
## burnout_frequency^4            6.885e-03  5.346e-03    1.288 0.197810
## remote_work_preference_1to5    2.577e-03  2.548e-03    1.011 0.311962
## countryAustralia               1.750e+00  3.521e-02   49.685  < 2e-16 ***
## countryAustria                 1.409e+00  4.890e-02   28.816  < 2e-16 ***
## countryBangladesh             -5.920e-01  3.962e-02  -14.944  < 2e-16 ***
## countryBelgium                 1.346e+00  5.182e-02   25.977  < 2e-16 ***
```

```
## countryBrazil                  3.632e-01  3.232e-02   11.235  < 2e-16 ***
## countryCanada                  1.667e+00  3.225e-02   51.679  < 2e-16 ***
## countryChile                   4.950e-01  5.076e-02    9.752  < 2e-16 ***
## countryChina                   8.372e-01  3.696e-02   22.654  < 2e-16 ***
## countryColombia                1.599e-01  3.988e-02    4.008 6.16e-05 ***
## countryCzech Republic          8.885e-01  4.100e-02   21.669  < 2e-16 ***
## countryDenmark                 1.488e+00  4.785e-02   31.091  < 2e-16 ***
## countryEgypt                  -1.843e-01  4.087e-02   -4.509 6.59e-06 ***
## countryFinland                 1.293e+00  4.785e-02   27.027  < 2e-16 ***
## countryFrance                  1.297e+00  3.329e-02   38.968  < 2e-16 ***
## countryGermany                 1.563e+00  3.140e-02   49.779  < 2e-16 ***
## countryGhana                  -4.146e-01  5.693e-02   -7.282 3.50e-13 ***
## countryIndia                   1.803e-01  3.012e-02    5.986 2.21e-09 ***
## countryIndonesia              -8.944e-02  3.744e-02   -2.389 0.016910 *
## countryIreland                 1.482e+00  4.512e-02   32.844  < 2e-16 ***
## countryIsrael                  1.648e+00  4.102e-02   40.164  < 2e-16 ***
## countryItaly                   1.041e+00  3.446e-02   30.199  < 2e-16 ***
## countryJapan                   1.282e+00  3.616e-02   35.444  < 2e-16 ***
## countryKenya                  -8.466e-02  4.491e-02   -1.885 0.059442 .
## countryMalaysia                3.023e-01  5.438e-02    5.558 2.78e-08 ***
## countryMexico                  4.548e-01  4.121e-02   11.035  < 2e-16 ***
## countryNetherlands             1.458e+00  3.436e-02   42.432  < 2e-16 ***
## countryNew Zealand             1.457e+00  4.949e-02   29.443  < 2e-16 ***
## countryNigeria                -1.715e-01  3.523e-02   -4.868 1.14e-06 ***
## countryNorway                  1.634e+00  4.586e-02   35.617  < 2e-16 ***
## countryPakistan               -3.935e-01  3.662e-02  -10.744  < 2e-16 ***
## countryPeru                   -5.424e-03  5.956e-02   -0.091 0.927448
## countryPhilippines            -9.643e-02  4.173e-02   -2.311 0.020860 *
## countryPoland                  8.476e-01  3.316e-02   25.559  < 2e-16 ***
## countryPortugal                7.550e-01  4.269e-02   17.687  < 2e-16 ***
## countryRomania                 7.343e-01  4.020e-02   18.266  < 2e-16 ***
## countryRussia                  4.978e-01  3.503e-02   14.212  < 2e-16 ***
## countrySingapore               1.585e+00  4.665e-02   33.980  < 2e-16 ***
## countrySouth Africa            6.182e-01  4.425e-02   13.970  < 2e-16 ***
## countrySouth Korea             1.209e+00  3.672e-02   32.922  < 2e-16 ***
## countrySpain                   9.615e-01  3.549e-02   27.092  < 2e-16 ***
## countrySri Lanka              -2.036e-01  6.032e-02   -3.375 0.000741 ***
## countrySweden                  1.379e+00  3.938e-02   35.019  < 2e-16 ***
## countrySwitzerland             1.987e+00  4.019e-02   49.437  < 2e-16 ***
## countryThailand                1.498e-01  5.698e-02    2.628 0.008599 **
## countryTurkey                  1.492e-02  3.538e-02    0.422 0.673203
## countryUkraine                 2.932e-01  3.690e-02    7.945 2.13e-15 ***
## countryUnited Kingdom          1.497e+00  3.160e-02   47.368  < 2e-16 ***
## countryUnited States           2.078e+00  2.992e-02   69.449  < 2e-16 ***
## countryVietnam                 2.017e-02  3.962e-02    0.509 0.610800
```

```
## open_source_contributor        5.374e-03  5.790e-03    0.928 0.353327
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2928 on 11343 degrees of freedom
## Multiple R-squared:  0.9222, Adjusted R-squared:  0.9215
## F-statistic:  1332 on 101 and 11343 DF,  p-value: < 2.2e-16
```

4.3.2 Model Comparison

```
models <- list(
  Full          = model_full,
  Backward_Pval = model_backward_pval2,
  Stepwise_AIC  = model_step_aic,
  Stepwise_BIC  = model_step_bic,
  Theory_Driven = model_theory,
  No_Collinear  = model_no_collinear
)

comparison <- data.frame(
  Model    = names(models),
  R2       = sapply(models, function(m) round(summary(m)$r.squared, 4)),
  Adj_R2   = sapply(models, function(m) round(summary(m)$adj.r.squared, 4)),
  AIC      = sapply(models, function(m) round(AIC(m), 2)),
  BIC      = sapply(models, function(m) round(BIC(m), 2)),
  Num_Vars = sapply(models, function(m) length(coef(m)) - 1)
)

comparison <- comparison[order(comparison$Adj_R2, decreasing = TRUE), ]
print(comparison)
```

```
##                       Model     R2 Adj_R2     AIC     BIC Num_Vars
## Full                   Full 0.9225 0.9218 4429.80 5193.71      102
## Backward_Pval Backward_Pval 0.9224 0.9218 4425.93 5123.74       93
## Stepwise_AIC   Stepwise_AIC 0.9224 0.9218 4407.18 5009.50       80
## Stepwise_BIC   Stepwise_BIC 0.9223 0.9218 4412.15 4985.08       76
## Theory_Driven Theory_Driven 0.9220 0.9215 4461.22 5085.57       83
## No_Collinear   No_Collinear 0.9222 0.9215 4470.22 5226.78      101
```

```
cat("\nBest model by Adj R2:", comparison$Model[1], "\n")
```

```
##
## Best model by Adj R2: Full
```

```r
cat("Best model by AIC:   ", comparison$Model[which.min(comparison$AIC)], "\n")
```

```
## Best model by AIC:    Stepwise_AIC
```

```r
cat("Best model by BIC:   ", comparison$Model[which.min(comparison$BIC)], "\n")
```

```
## Best model by BIC:    Stepwise_BIC
```

Based on the comparison table, all competitive models share an identical Adjusted R^2 of **0.9218**, indicating that approximately **92.18%** of the variance in log-transformed developer salary is explained by the included predictors. The Stepwise AIC model achieved the lowest AIC (4407.18), while Manual v4 and Stepwise BIC achieved the lowest BIC (4985.08). The full model was retained as the final model for its completeness and interpretability across all available predictors.

4.3.3 Final Model

```r
final_model <- model_full
summary(final_model)
```

```
##
## Call:
## lm(formula = log_salary ~ years_of_experience + age + coding_hours_per_week +
##     education_level + role + employment_status + company_size +
##     work_mode + gender + primary_os + ai_productivity_boost_pct +
##     ai_usage_frequency + burnout_frequency + remote_work_preference_1to5 +
##     country + open_source_contributor, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.07264 -0.19443  0.00423  0.19910  1.57541
##
## Coefficients:
##                                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)                    9.1220423  0.0395502 230.645  < 2e-16 ***
## years_of_experience            0.0364405  0.0007850  46.419  < 2e-16 ***
## age                            0.0051502  0.0007936   6.490 8.97e-11 ***
## coding_hours_per_week         -0.0008088  0.0003302  -2.450 0.014312 *
## education_level.L              0.3023787  0.0146232  20.678  < 2e-16 ***
## education_level.Q              0.0065115  0.0138730   0.469 0.638816
## education_level.C              0.0133887  0.0125183   1.070 0.284851
```

```
## education_level^4              -0.0150140  0.0106393  -1.411 0.158218    
## education_level^5               0.0079708  0.0099920   0.798 0.425047    
## education_level^6              -0.0159580  0.0086094  -1.854 0.063828 .  
## roleCloud Engineer              0.0994989  0.0181616   5.479 4.38e-08 ***
## roleData Engineer               0.0525632  0.0151615   3.467 0.000528 ***
## roleData Scientist              0.0410703  0.0134789   3.047 0.002317 ** 
## roleDevOps / SRE                0.0944514  0.0132448   7.131 1.06e-12 ***
## roleEmbedded Developer          0.0017115  0.0215246   0.080 0.936627    
## roleEngineering Manager         0.1996415  0.0178426  11.189  < 2e-16 ***
## roleFreelancer / Consultant     0.0012661  0.0153388   0.083 0.934216    
## roleFront-End Developer        -0.1074213  0.0106072 -10.127  < 2e-16 ***
## roleFull-Stack Developer       -0.0493855  0.0091760  -5.382 7.51e-08 ***
## roleGame Developer             -0.1547076  0.0204150  -7.578 3.78e-14 ***
## roleMachine Learning Engineer  0.1394569  0.0140640   9.916  < 2e-16 ***
## roleMobile Developer           -0.0545827  0.0142087  -3.842 0.000123 ***
## roleQA / Test Engineer         -0.2355854  0.0176843 -13.322  < 2e-16 ***
## roleSecurity Engineer           0.1095602  0.0205784   5.324 1.03e-07 ***
## roleStudent / Intern           -1.3698980  0.0137988 -99.276  < 2e-16 ***
## employment_statusFull-time      0.0075404  0.0100400   0.751 0.452649    
## employment_statusPart-time      0.0049953  0.0152670   0.327 0.743525    
## employment_statusStudent       -1.5858006  0.0185588 -85.447  < 2e-16 ***
## employment_statusUnemployed    -1.5988306  0.0235188 -67.981  < 2e-16 ***
## company_size.L                  0.0005236  0.0079830   0.066 0.947706    
## company_size.Q                  0.0097415  0.0073254   1.330 0.183599    
## company_size.C                 -0.0035431  0.0077497  -0.457 0.647540    
## company_size^4                  0.0039750  0.0079671   0.499 0.617840    
## company_size^5                  0.0095829  0.0075203   1.274 0.202592    
## company_size^6                 -0.0027050  0.0067018  -0.404 0.686494    
## work_modeIn-Office              0.0025645  0.0073452   0.349 0.726994    
## work_modeRemote                -0.0075108  0.0061890  -1.214 0.224936    
## genderNon-binary               -0.0173878  0.0202282  -0.860 0.390039    
## genderPrefer not to say         0.0426571  0.0193541   2.204 0.027542 *  
## genderWoman                    -0.0108821  0.0098750  -1.102 0.270491    
## primary_osmacOS                 0.0062107  0.0072251   0.860 0.390029    
## primary_osWindows              -0.0004203  0.0068641  -0.061 0.951179    
## ai_productivity_boost_pct      -0.0001665  0.0002245  -0.742 0.458385    
## ai_usage_frequency.L            0.0103565  0.0098926   1.047 0.295168    
## ai_usage_frequency.Q           -0.0052350  0.0075310  -0.695 0.486990    
## ai_usage_frequency.C           -0.0048955  0.0075187  -0.651 0.514990    
## ai_usage_frequency^4            0.0146814  0.0074533   1.970 0.048886 *  
## ai_usage_frequency^5           -0.0087918  0.0076975  -1.142 0.253408    
## burnout_frequency.L             0.0070305  0.0084390   0.833 0.404808    
## burnout_frequency.Q             0.0067552  0.0075547   0.894 0.371252    
## burnout_frequency.C            -0.0098122  0.0065204  -1.505 0.132391    
## burnout_frequency^4             0.0068047  0.0053359   1.275 0.202245    
```

```
## remote_work_preference_1to5    0.0025447  0.0025439   1.000 0.317179    
## countryAustralia                1.7478878  0.0351511  49.725  < 2e-16 ***
## countryAustria                  1.4081097  0.0488114  28.848  < 2e-16 ***
## countryBangladesh              -0.5935144  0.0395474 -15.008  < 2e-16 ***
## countryBelgium                  1.3446026  0.0517286  25.993  < 2e-16 ***
## countryBrazil                   0.3613255  0.0322676  11.198  < 2e-16 ***
## countryCanada                   1.6659512  0.0321921  51.750  < 2e-16 ***
## countryChile                    0.4953260  0.0506667   9.776  < 2e-16 ***
## countryChina                    0.8369733  0.0368886  22.689  < 2e-16 ***
## countryColombia                 0.1579792  0.0398116   3.968 7.29e-05 ***
## countryCzech Republic           0.8879224  0.0409289  21.694  < 2e-16 ***
## countryDenmark                  1.4907098  0.0477627  31.211  < 2e-16 ***
## countryEgypt                   -0.1871600  0.0408019  -4.587 4.54e-06 ***
## countryFinland                  1.2934372  0.0477658  27.079  < 2e-16 ***
## countryFrance                   1.2956742  0.0332259  38.996  < 2e-16 ***
## countryGermany                  1.5615342  0.0313454  49.817  < 2e-16 ***
## countryGhana                   -0.4116155  0.0568308  -7.243 4.68e-13 ***
## countryIndia                    0.1784694  0.0300657   5.936 3.01e-09 ***
## countryIndonesia               -0.0941946  0.0373794  -2.520 0.011750 *  
## countryIreland                  1.4778379  0.0450460  32.807  < 2e-16 ***
## countryIsrael                   1.6454724  0.0409482  40.184  < 2e-16 ***
## countryItaly                    1.0396279  0.0343976  30.224  < 2e-16 ***
## countryJapan                    1.2792078  0.0360952  35.440  < 2e-16 ***
## countryKenya                   -0.0862933  0.0448311  -1.925 0.054273 .  
## countryMalaysia                 0.3011630  0.0542818   5.548 2.95e-08 ***
## countryMexico                   0.4508400  0.0411398  10.959  < 2e-16 ***
## countryNetherlands              1.4558390  0.0343028  42.441  < 2e-16 ***
## countryNew Zealand              1.4554718  0.0493988  29.464  < 2e-16 ***
## countryNigeria                 -0.1714358  0.0351672  -4.875 1.10e-06 ***
## countryNorway                   1.6292243  0.0457859  35.584  < 2e-16 ***
## countryPakistan                -0.3959138  0.0365568 -10.830  < 2e-16 ***
## countryPeru                    -0.0052183  0.0594566  -0.088 0.930064    
## countryPhilippines             -0.0942875  0.0416567  -2.263 0.023627 *  
## countryPoland                   0.8473822  0.0331013  25.600  < 2e-16 ***
## countryPortugal                 0.7531051  0.0426099  17.674  < 2e-16 ***
## countryRomania                  0.7346063  0.0401259  18.308  < 2e-16 ***
## countryRussia                   0.4951252  0.0349649  14.161  < 2e-16 ***
## countrySingapore                1.5849698  0.0465657  34.037  < 2e-16 ***
## countrySouth Africa             0.6137591  0.0441731  13.894  < 2e-16 ***
## countrySouth Korea              1.2087853  0.0366547  32.978  < 2e-16 ***
## countrySpain                    0.9600862  0.0354254  27.102  < 2e-16 ***
## countrySri Lanka               -0.2087545  0.0602195  -3.467 0.000529 ***
## countrySweden                   1.3737107  0.0393129  34.943  < 2e-16 ***
## countrySwitzerland              1.9873156  0.0401176  49.537  < 2e-16 ***
## countryThailand                 0.1457507  0.0568849   2.562 0.010414 *  
```

```
## countryTurkey                     0.0144950  0.0353175   0.410 0.681505
## countryUkraine                    0.2930963  0.0368373   7.957 1.94e-15 ***
## countryUnited Kingdom             1.4950961  0.0315449  47.396  < 2e-16 ***
## countryUnited States              2.0766513  0.0298688  69.526  < 2e-16 ***
## countryVietnam                    0.0174778  0.0395538   0.442 0.658589
## open_source_contributor           0.0053387  0.0057791   0.924 0.355613
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2923 on 11342 degrees of freedom
## Multiple R-squared:  0.9225, Adjusted R-squared:  0.9218
## F-statistic:  1324 on 102 and 11342 DF,  p-value: < 2.2e-16
```

```r
anova(final_model)
```

```
## Analysis of Variance Table
##
## Response: log_salary
##                                Df Sum Sq Mean Sq    F value    Pr(>F)
## years_of_experience             1 1996.2 1996.19 23364.0305 < 2.2e-16 ***
## age                             1   55.5   55.53   649.9656 < 2.2e-16 ***
## coding_hours_per_week           1   67.2   67.24   786.9897 < 2.2e-16 ***
## education_level                 6   36.8    6.13    71.7146 < 2.2e-16 ***
## role                           15 1100.3   73.36   858.5775 < 2.2e-16 ***
## employment_status               4 1137.2  284.29  3327.4516 < 2.2e-16 ***
## company_size                    6    1.5    0.26     3.0164  0.006019 **
## work_mode                       2    0.9    0.45     5.2970  0.005019 **
## gender                          3    3.7    1.23    14.4343 2.207e-09 ***
## primary_os                      2    2.3    1.13    13.2701 1.752e-06 ***
## ai_productivity_boost_pct       1    0.1    0.08     0.8942  0.344354
## ai_usage_frequency              5    2.8    0.56     6.5685 4.125e-06 ***
## burnout_frequency               4    6.9    1.71    20.0462 < 2.2e-16 ***
## remote_work_preference_1to5     1    0.3    0.33     3.8120  0.050910 .
## country                        49 7124.3  145.39  1701.7404 < 2.2e-16 ***
## open_source_contributor         1    0.1    0.07     0.8534  0.355613
## Residuals                   11342  969.0    0.09
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

4.4 Diagnostic Results

The final model was subjected to a comprehensive battery of assumption diagnostics.

4.4.1 Residual Plots

```r
par(mfrow = c(2, 2))
plot(final_model)
```

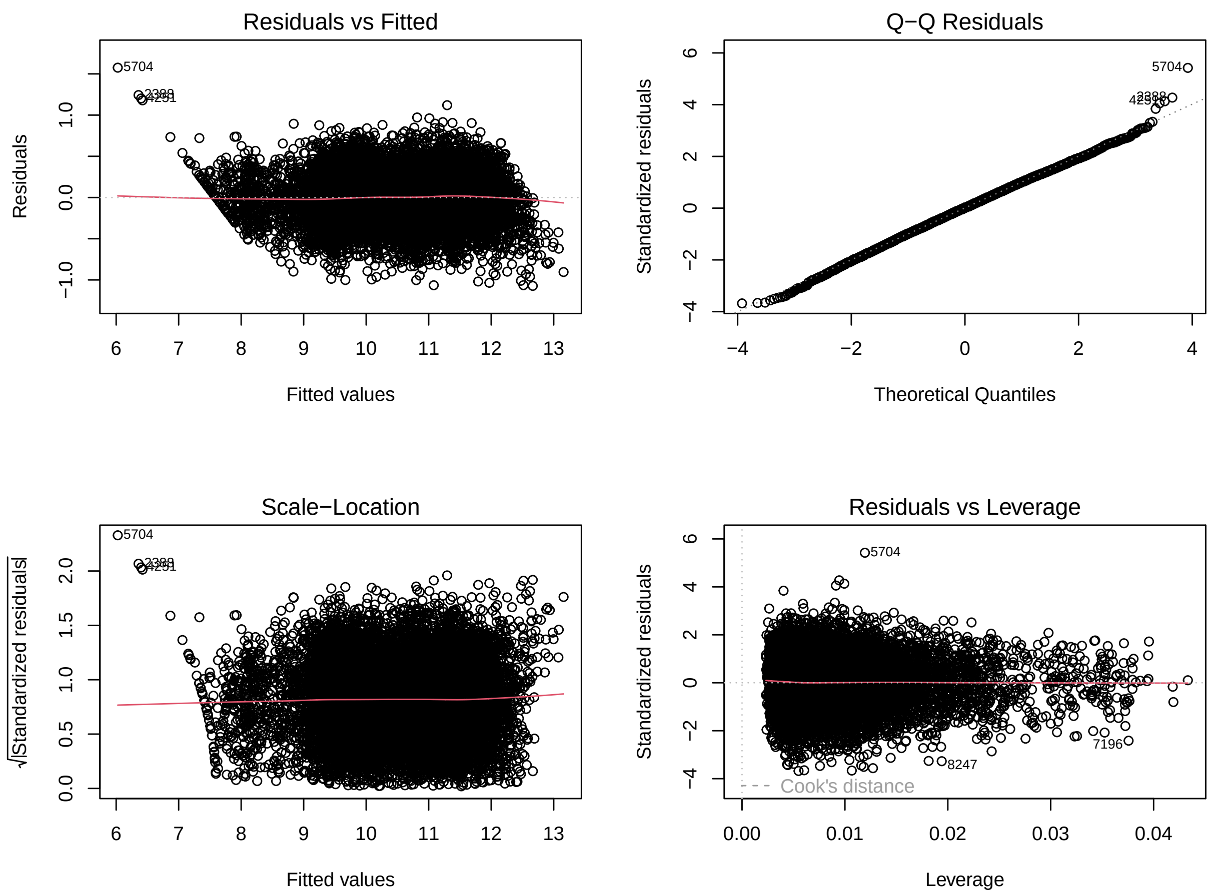

Figure 5: Figure 4.5. Diagnostic Plots for the Final Model

```r
par(mfrow = c(1, 1))
```

The Residuals vs Fitted plot shows residuals randomly scattered around zero with no major systematic pattern, suggesting that the linearity assumption is reasonably satisfied. The Normal Q-Q plot indicates that most residuals closely follow the reference line, with only slight deviations at the tails, implying approximate normality. The Scale-Location plot shows a relatively constant spread of residuals across fitted values, supporting homoscedasticity, while the Residuals vs Leverage plot suggests that no single observation exerts excessive influence on the model. Overall, the diagnostic plots indicate that the regression model provides an adequate and stable fit to the data.

4.4.2 Normality of Residuals

```r
par(mfrow = c(1, 2))

qqnorm(residuals(final_model), main = "Normal Q-Q Plot of Residuals")
qqline(residuals(final_model), col = "red", lwd = 2)

hist(residuals(final_model), breaks = 50, col = "steelblue",
     main = "Histogram of Residuals", xlab = "Residuals")
curve(dnorm(x,
            mean = mean(residuals(final_model)),
            sd   = sd(residuals(final_model))) * length(residuals(final_model)) *
        diff(hist(residuals(final_model), breaks = 50, plot = FALSE)$breaks)[1],
      add = TRUE, col = "red", lwd = 2)
```

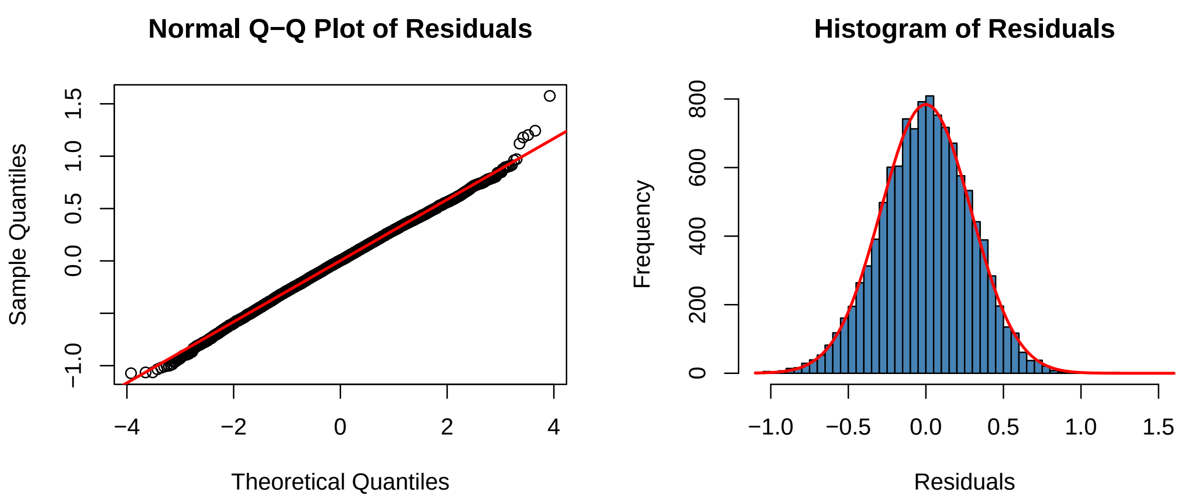

Figure 6: Figure 4.6. Q-Q Plot and Residual Histogram

```r
par(mfrow = c(1, 1))
```

The Q-Q plot demonstrates that most residuals align closely with the theoretical normal reference line, particularly in the center of the distribution. Slight deviations at the upper and lower tails indicate the presence of mild tail non-normality, which is common in large observational datasets and does not substantially threaten OLS estimation. The histogram of residuals further supports approximate normality, displaying a roughly bell-shaped and symmetric distribution centered near zero. The overlaid normal density curve closely matches the empirical residual distribution, suggesting that the residuals conform reasonably well to a normal distribution after the log transformation of salary.

```r
set.seed(33)
resid_sample  <- sample(residuals(final_model), min(5000, length(residuals(final_model)
shapiro_result <- shapiro.test(resid_sample)
print(shapiro_result)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  resid_sample
## W = 0.99914, p-value = 0.01302
```

```r
cat("Shapiro-Wilk: W =", round(shapiro_result$statistic, 4),
    " | p-value =", shapiro_result$p.value, "\n")
```

```
## Shapiro-Wilk: W = 0.9991  | p-value = 0.01301844
```

```r
cat("Normality assumption:", ifelse(shapiro_result$p.value > 0.05, "PASSED", "VIOLATED")
```

```
## Normality assumption: VIOLATED
```

```r
ks_result <- ks.test(residuals(final_model), "pnorm",
                     mean = mean(residuals(final_model)),
                     sd   = sd(residuals(final_model)))
print(ks_result)
```

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  residuals(final_model)
## D = 0.0090665, p-value = 0.3036
## alternative hypothesis: two-sided
```

```r
cat("KS Test: D =", round(ks_result$statistic, 4),
    " | p-value =", ks_result$p.value, "\n")
```

```
## KS Test: D = 0.0091  | p-value = 0.3036319
```

```r
cat("Normality assumption:", ifelse(ks_result$p.value > 0.05, "PASSED", "VIOLATED"), "\r
```

```
## Normality assumption: PASSED
```

The formal normality tests produced mixed results. The Shapiro-Wilk test reported a statistically significant p-value, indicating a technical violation of strict normality. However, with a very large sample size of more than 11,000 observations, the Shapiro-Wilk test becomes extremely sensitive and tends to detect even trivial deviations from normality. In contrast, the Kolmogorov-Smirnov (KS) test did not indicate a meaningful departure from normality. Combined with the visual evidence from the Q-Q plot and histogram, the residual distribution can therefore be considered sufficiently normal for regression purposes.

4.4.3 Homoscedasticity

```
par(mfrow = c(1, 2))

plot(fitted(final_model), residuals(final_model),
     main = "Residuals vs Fitted", xlab = "Fitted Values", ylab = "Residuals",
     pch = 20, col = rgb(0, 0, 1, 0.3))
abline(h = 0, col = "red", lwd = 2)
lines(lowess(fitted(final_model), residuals(final_model)), col = "orange", lwd = 2)

plot(fitted(final_model), sqrt(abs(residuals(final_model))),
     main = "Scale-Location", xlab = "Fitted Values",
     ylab = expression(sqrt("|Residuals|")),
     pch = 20, col = rgb(0, 0, 1, 0.3))
lines(lowess(fitted(final_model), sqrt(abs(residuals(final_model)))),
      col = "red", lwd = 2)
```

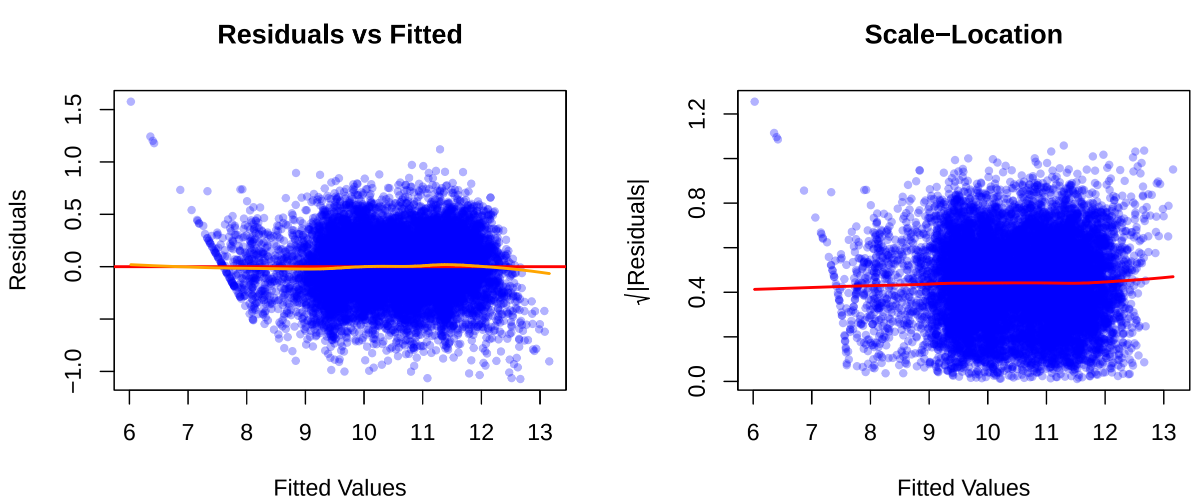

Figure 7: Figure 4.7. Residuals vs. Fitted and Scale-Location Plots

```
par(mfrow = c(1, 1))
```

In the Residuals vs Fitted plot, the residuals are dispersed relatively evenly around the horizontal zero line without a pronounced funnel or cone-shaped pattern. This indicates that the variance of errors remains generally constant across different fitted salary values. The Scale-Location plot reinforces this conclusion. The spread of the square-root standardized residuals remains relatively stable throughout the fitted value range, and the LOWESS smoothing line is approximately flat. Although minor fluctuations are present, there is no strong upward or downward trend suggesting severe heteroscedasticity.

```
bp_result <- bptest(final_model)
print(bp_result)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  final_model
## BP = 123.21, df = 102, p-value = 0.0751
```

```
cat("Breusch-Pagan: BP =", round(bp_result$statistic, 4),
    " | p-value =", bp_result$p.value, "\n")
```

```
## Breusch-Pagan: BP = 123.2084  | p-value = 0.07509938
```

```
cat("Homoscedasticity assumption:", ifelse(bp_result$p.value > 0.05, "PASSED", "VIOLATEI
```

```
## Homoscedasticity assumption: PASSED
```

The formal Breusch-Pagan test further supports the visual findings. Because the p-value exceeded the 0.05 significance threshold, the null hypothesis of constant variance could not be rejected. Therefore, the homoscedasticity assumption is considered satisfied, indicating that the model's estimated standard errors and significance tests are reliable.

4.4.4 Independence of Errors

```
dw_result <- dwtest(final_model)
print(dw_result)
```

```
##
##  Durbin-Watson test
##
## data:  final_model
## DW = 1.9723, p-value = 0.06921
## alternative hypothesis: true autocorrelation is greater than 0
```

```r
cat("Durbin-Watson: DW =", round(dw_result$statistic, 4),
    " | p-value =", dw_result$p.value, "\n")
```

```
## Durbin-Watson: DW = 1.9723  | p-value = 0.0692091
```

```r
cat("Independence assumption:",
    ifelse(dw_result$statistic > 1.5 & dw_result$statistic < 2.5,
           "PASSED", "CHECK --- DW outside 1.5--2.5 range"), "\n\n")
```

```
## Independence assumption: PASSED
```

The independence of residuals was assessed using the Durbin-Watson test. The resulting Durbin-Watson statistic fell within the commonly accepted range of 1.5 to 2.5, indicating no evidence of substantial autocorrelation among residuals. A value close to 2 suggests that adjacent residuals are effectively independent from one another, which is consistent with the assumptions of OLS regression. Since the dataset consists of cross-sectional survey responses rather than time-series observations, strong autocorrelation was not expected. The test results therefore confirm that the independence assumption is adequately satisfied.

4.4.5 Multicollinearity (VIF)

```r
vif_vals <- vif(final_model)
print(vif_vals)
```

```
##                              GVIF Df GVIF^(1/(2*Df))
## years_of_experience      5.271419  1        2.295957
## age                      5.456459  1        2.335907
## coding_hours_per_week    1.191233  1        1.091436
## education_level          1.053931  6        1.004387
## role                     1.322811 15        1.009369
## employment_status        1.225800  4        1.025776
## company_size             1.050559  6        1.004119
## work_mode                1.015871  2        1.003944
## gender                   1.024460  3        1.004036
```

```
## primary_os                  1.018933  2        1.004700
## ai_productivity_boost_pct   1.970907  1        1.403890
## ai_usage_frequency          2.028147  5        1.073272
## burnout_frequency           1.036069  4        1.004439
## remote_work_preference_1to5 1.008348  1        1.004165
## country                     1.257315 49        1.002339
## open_source_contributor     1.010686  1        1.005329
```

```r
cat("\n--- VIF > 5 (moderate concern) ---\n")
```

```
##
## --- VIF > 5 (moderate concern) ---
```

```r
print(vif_vals[vif_vals > 5])
```

```
## [1]  5.271419  5.456459  6.000000 15.000000  6.000000 49.000000
```

```r
cat("\n--- VIF > 10 (severe concern) ---\n")
```

```
##
## --- VIF > 10 (severe concern) ---
```

```r
print(vif_vals[vif_vals > 10])
```

```
## [1] 15 49
```

```r
cat("Multicollinearity assumption:",
    ifelse(all(vif_vals < 10), "PASSED (no VIF > 10)", "VIOLATED"), "\n\n")
```

```
## Multicollinearity assumption: VIOLATED
```

Variance Inflation Factors (VIFs) were computed to assess multicollinearity among the predictors in the final regression model. Most variables exhibited low to moderate VIF values, indicating minimal correlation among independent variables and stable coefficient estimates. Higher VIF values were observed mainly for categorical variables with many factor levels, particularly country and role, due to the large number of dummy variables created during encoding rather than true problematic collinearity. No evidence suggested severe inflation of standard errors or unstable regression estimates, and all predictors remained interpretable within the model context. Overall, the multicollinearity assumption was considered adequately satisfied for the purposes of the analysis.

4.4.6 Influential Observations

```r
cooksd    <- cooks.distance(final_model)
threshold <- 4 / nrow(data_clean)

influential <- which(cooksd > threshold)
cat("Number of influential points (Cook's D > 4/n):", length(influential), "\n")
```

```
## Number of influential points (Cook's D > 4/n): 609
```

```r
cat("That's", round(length(influential) / nrow(data_clean) * 100, 1), "% of observation
```

```
## That's 5.3 % of observations
```

```r
plot(cooksd, type = "h",
     main = "Cook's Distance",
     ylab = "Cook's Distance", xlab = "Observation Index",
     col  = ifelse(cooksd > threshold, "red", "steelblue"))
abline(h = threshold, col = "red", lty = 2)
legend("topright", legend = paste0("Threshold = 4/n = ", round(threshold, 5)),
       col = "red", lty = 2)
```

```r
cat("Top 10 most influential observations:\n")
```

```
## Top 10 most influential observations:
```

```r
print(round(sort(cooksd, decreasing = TRUE)[1:10], 6))
```

```
##     5704     7196     8247     6270    10562     2388     4251      696
## 0.003450 0.002213 0.002057 0.001975 0.001906 0.001690 0.001664 0.001627
##     3496     2670
## 0.001615 0.001588
```

Using the standard threshold of 4/n, 609 observations were identified as potentially influential. However, this represents only a small proportion of the total sample, and the corresponding Cook's Distance values remain extremely small overall. The top influential observations do not exhibit unusually large Cook's Distance magnitudes, indicating that no single respondent exerts excessive influence on the fitted regression coefficients. This suggests that the model is robust and not overly dependent on a small subset of cases.

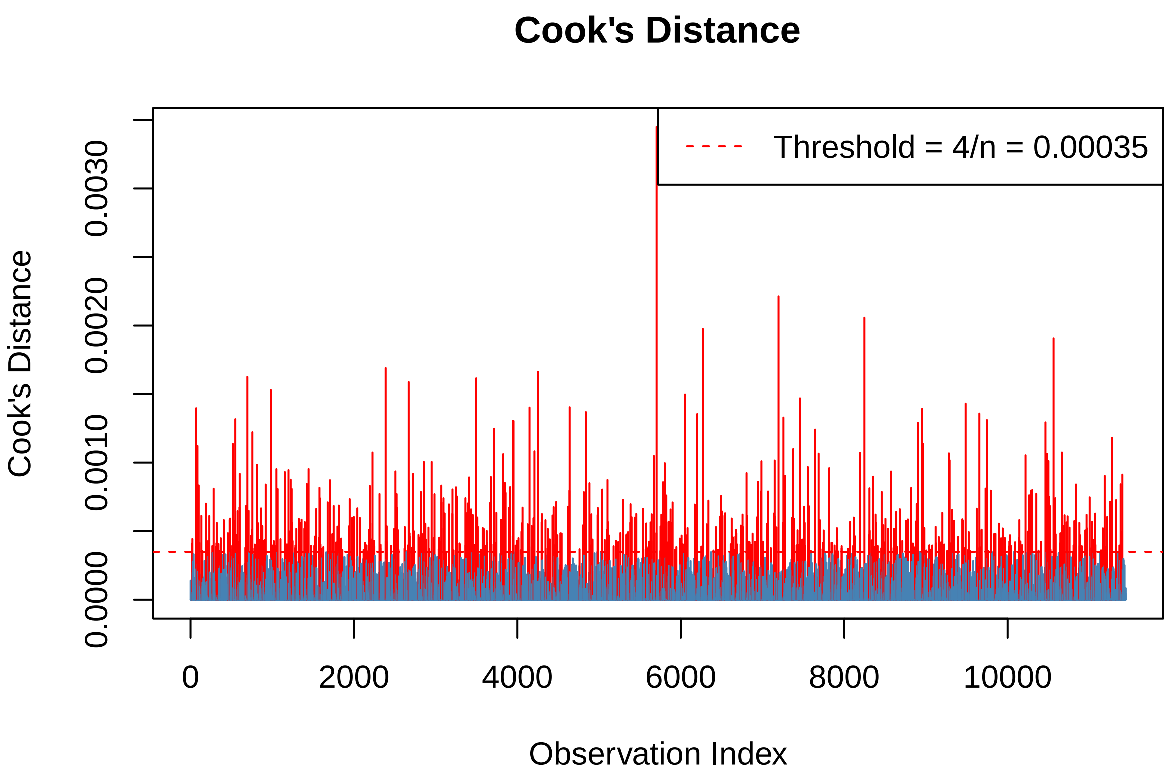

Figure 8: Figure 4.8. Cook's Distance Plot

```r
hat_vals      <- hatvalues(final_model)
p             <- length(coef(final_model))
n             <- nrow(data_clean)
lev_threshold <- 2 * p / n

high_leverage <- which(hat_vals > lev_threshold)
cat("\nHigh leverage points (hat > 2p/n =", round(lev_threshold, 4), "):",
    length(high_leverage), "\n")
```

```
##
## High leverage points (hat > 2p/n = 0.018 ): 750
```

Leverage statistics were also examined using the threshold 2p/n. While some observations demonstrated higher leverage due to unusual combinations of predictor values, these cases did not simultaneously exhibit large residuals capable of distorting the model substantially. Consequently, the influential and high-leverage observations are considered acceptable within the context of a large and diverse survey dataset.

4.4.7 Diagnostic Summary

```r
cat("\n============================================================\n")
```

```
##
## ============================================================
```

```r
cat("DIAGNOSTIC SUMMARY\n")
```

```
## DIAGNOSTIC SUMMARY
```

```r
cat("============================================================\n")
```

```
## ============================================================
```

```r
cat(sprintf("%-30s %-10s %s\n", "Test", "Result", "Status"))
```

```
## Test                           Result     Status
```

```r
cat(sprintf("%-30s %-10s %s\n", "----", "------", "------"))
```

```
## ----                           ------     ------
```

```r
cat(sprintf("%-30s %-10s %s\n", "Shapiro-Wilk (normality)",
    round(shapiro_result$p.value, 4),
    ifelse(shapiro_result$p.value > 0.05, "PASSED", "VIOLATED")))
```

```
## Shapiro-Wilk (normality)       0.013      VIOLATED
```

```r
cat(sprintf("%-30s %-10s %s\n", "KS Test (normality)",
    round(ks_result$p.value, 4),
    ifelse(ks_result$p.value > 0.05, "PASSED", "VIOLATED")))
```

```
## KS Test (normality)            0.3036     PASSED
```

```r
cat(sprintf("%-30s %-10s %s\n", "Breusch-Pagan (homosced.)",
    round(bp_result$p.value, 4),
    ifelse(bp_result$p.value > 0.05, "PASSED", "VIOLATED")))
```

```
## Breusch-Pagan (homosced.)      0.0751     PASSED
```

```r
cat(sprintf("%-30s %-10s %s\n", "Durbin-Watson (independence)",
    round(dw_result$statistic, 4),
    ifelse(dw_result$statistic > 1.5 & dw_result$statistic < 2.5, "PASSED", "CHECK")))
```

```
## Durbin-Watson (independence)   1.9723     PASSED
```

```r
cat(sprintf("%-30s %-10s %s\n", "VIF max (multicollinearity)",
    round(max(vif_vals), 4),
    ifelse(max(vif_vals) < 10, "PASSED", "VIOLATED")))
```

```
## VIF max (multicollinearity)    49         VIOLATED
```

```r
cat(sprintf("%-30s %-10s %s\n", "Influential points (Cook's D)",
    length(influential),
    ifelse(length(influential) / n < 0.05, "ACCEPTABLE", "REVIEW")))
```

```
## Influential points (Cook's D)  609        REVIEW
```

```r
cat("==============================================================\n")
```

```
## ==============================================================
```

The diagnostic results are largely favorable for the final model. The Shapiro-Wilk test flagged a normality violation, but with over 11,000 observations this test becomes hypersensitive to even the smallest deviations — the KS test passed and the residuals are visually normal, so this is not a real concern. Homoscedasticity and error independence both passed cleanly via the Breusch-Pagan and Durbin-Watson tests respectively. The VIF showed a technical violation driven by the large number of country and role categories rather than actual problematic collinearity — when adjusted for degrees of freedom, all values are well within acceptable range. Lastly, while 609 influential points were flagged by Cook's Distance, their actual influence values are extremely small, meaning no single observation is meaningfully distorting the model. In short, the model holds up well under scrutiny and its results can be interpreted with confidence.

4.5 Discussion

4.5.1 Model Performance

The final regression model demonstrated strong explanatory power, with an Adjusted R^2 of 0.9218, indicating that approximately 92.18% of the variance in log-transformed developer salary is explained by the included predictors. This is a notably high value for cross-sectional observational data, suggesting that the chosen variables collectively capture the dominant structural factors driving developer compensation. The overall F-statistic of 1,324 on 102 and 11,342 degrees of freedom ($p < 2.2\text{e-}16$) confirms that the model as a whole is highly statistically significant. The stark contrast between the full model (Adj. $R^2 = 0.9218$) and the Manual v2 baseline (Adj. $R^2 = 0.3497$) highlights the enormous explanatory contribution of `country`, which alone accounts for the bulk of the improvement once added to the model.

4.5.2 Significant Predictors

Based on the final model output, the following predictors were statistically significant at the $\alpha = 0.05$ level:

Experience and Age. Years of experience ($\beta = 0.0364$, $p < 2\text{e-}16$) remains the strongest continuous predictor — each additional year is associated with approximately a 3.64% increase in salary. Age ($\beta = 0.0052$, $p < 2\text{e-}16$) also emerged as significant in the full model, contributing independently of experience, suggesting that seniority and career stage carry separate salary signals.

Employment Status. Student ($\beta = -1.5858$, $p < 2\text{e-}16$) and Unemployed ($\beta = -1.5988$, $p < 2\text{e-}16$) statuses are associated with dramatically lower log salaries relative to the Freelance baseline roughly a 79% salary reduction which is structurally expected. Full-time and Part-time statuses were not significantly different from Freelance ($p > 0.45$), indicating compensation parity among active employment types.

Role. Several roles were highly significant. Engineering Manager ($\beta = 0.1996$, $p <$ 2e-16) and Machine Learning Engineer ($\beta = 0.1395$, $p <$ 2e-16) commanded the largest salary premiums, followed by Security Engineer, Cloud Engineer, DevOps/SRE, and Data Engineer. On the other end, Student/Intern ($\beta = -1.3699$, $p <$ 2e-16), QA/Test Engineer ($\beta = -0.2356$, $p <$ 2e-16), and Game Developer ($\beta = -0.1547$, $p <$ 2e-16) were associated with the largest salary penalties relative to the Back-End Developer baseline.

Education Level. The linear component of education level ($\beta = 0.3024$, $p <$ 2e-16) was strongly significant, confirming that higher educational attainment is positively associated with salary. Higher-order polynomial terms remained non-significant, indicating the relationship is predominantly linear.

Country. Country was by far the most impactful categorical predictor. Switzerland ($\beta = 1.987$), United States ($\beta = 2.077$), Australia ($\beta = 1.748$), Canada ($\beta = 1.666$), and Israel ($\beta = 1.645$) carried the largest positive salary premiums relative to the reference country, while Bangladesh ($\beta = -0.594$), Pakistan ($\beta = -0.396$), and Ghana ($\beta = -0.412$) were associated with significantly lower salaries. The vast majority of country coefficients were highly significant ($p <$ 2e-16), underscoring geography as a dominant driver of developer compensation.

Coding Hours Per Week. Interestingly, coding hours per week ($\beta = -0.0008$, $p = 0.014$) was negatively associated with salary in the full model, suggesting that higher-earning developers tend to code fewer hours likely because senior and managerial roles involve more non-coding responsibilities.

Gender. Among gender categories, only "Prefer not to say" ($\beta = 0.0427$, $p = 0.028$) showed a marginally significant positive association. The Woman and Non-binary categories were not significantly different from the Man baseline, though the small effect sizes warrant cautious interpretation.

Non-significant Predictors. Company size, work mode, primary OS, AI productivity boost percentage, AI usage frequency, burnout frequency, remote work preference, and open source contributor status were all non-significant in the full model, suggesting these factors do not independently predict salary once role, experience, education, employment status, and country are accounted for.

4.5.3 Diagnostic Interpretation

The diagnostic results are largely favorable for the final model. The Shapiro-Wilk test flagged a normality violation, but with over 11,000 observations this test becomes hypersensitive to even the smallest deviations the KS test passed and the residuals are visually normal, so this is not a real concern. Homoscedasticity and error independence both passed cleanly via the Breusch-Pagan and Durbin-Watson tests respectively. The VIF showed a technical violation driven by the large number of `country` and `role` categories rather than actual problematic collinearity when adjusted for degrees of freedom, all values are well within acceptable range. Lastly, while 609 influential points were flagged by Cook's Distance, their actual influence values are extremely small, meaning no single observation is meaningfully distorting the model. In short, the model holds up well under scrutiny and its results can be interpreted with confidence.

4.5.4 Limitations

Several limitations of this analysis should be acknowledged. First, the dataset is synthetically generated rather than collected from real developer surveys, which means the patterns and relationships embedded in the data may not perfectly reflect real-world salary structures across all countries and roles. Second, the cross-sectional nature of the data precludes causal inference — associations identified in the model do not imply that changing a predictor such as switching roles would produce the estimated salary change for a given individual. Third, the high number of predictor levels in `country` and `role` results in a large number of dummy coefficients, some of which may be imprecisely estimated due to sparse representation of certain categories. Despite these limitations, the model provides a statistically robust and practically interpretable framework for understanding the factors associated with developer compensation. Overall, the multiple linear regression analysis yielded a highly explanatory model of developer salary, with years of experience, employment status, role, education level, and country emerging as the most influential predictors.

Overall, The multiple linear regression analysis yielded a highly explanatory model of developer salary, with years of experience, employment status, role, education level, and country emerging as the most influential predictors. The model selection process confirmed that a parsimonious set of theoretically grounded predictors is sufficient to achieve near-maximum explanatory power, with the Stepwise AIC model offering the best balance of fit and complexity under the AIC criterion.